Disease transmission models using Survival Analysis

Research Proposal

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Contents

1 Introduction **4**

1.1 Modelling Tuberculosis transmission in British Columbia 5

1.2 Research Questions 7

2 Literature Review **8**

2.1 Reconstructing transmission trees 8

2.1.1 Adding phylogenetic trees 8

2.1.2 Survival Analysis in disease transmission 11

2.1.3 Spatial data in disease transmission 11

3 Phylogenetic Trees **13**

3.1 Inferring Phylogenetic trees 14

3.1.1 Number of trees and branch lengths of rooted trees 14

3.1.2 Parsimony and Distance Methods 16

3.1.3 Likelihood-based methods 17

3.2 Likelihood method for phylogenetic trees 18

3.2.1 Substitution models 18

3.2.2 Likelihood of a phylogeny 20

3.3 Bayesian methods for phylogenetic trees 23

3.3.1 Population dynamics model 23

3.3.2 Likelihood of observed sequence 24

3.4 Software for Phylogenetic Trees 26

4 SEIR and Survival Framework **26**

4.1 Survival Analysis for transmission data 27

4.2 Likelihood for a single transmission tree 28

4.3 Likelihood for multiple transmission trees 29

4.4 Accounting for covariates in Epidemiological models 29

5 Expectation-Maximization Algorithm for Parameter Estimation **30**

5.1 Introduction 30

5.2 Expectation Maximization (EM) Algorithm 31

5.2.1 The Expectation Maximization (EM) Algorithm 32

5.2.2 Applying variant to Standard EM algorithm 33

5.3 Monte Carlo Methods 35

5.3.1 Importance Sampling 35

5.3.2 Markov Chain Monte Carlo methods 36

5.4 Monte Carlo - EM 38

5.5 Variational approximation EM Algorithm 39

5.6 Accounting for uncertainty in phylogenetic trees for a frequentist framework 41

5.6.1 Monte Carlo-EM algorithm for Transmission model 41

5.6.2 The models and likelihood 42

5.6.3 Algorithm 43

6 Implementation **44**

7 Application **45**

7.1 Hazard and Survival Function 45

7.2 Results 46

7.3 Simulation Study 47

8 Further work **48**

1 Introduction

Infectious diseases that spread over a region at a quick rate may have a great impact on the socio-economic standing of the affected area. Loss of human resources cannot easily be replaced, making it important that the right measures are put in place to deal with such unwanted occurrences. Infectious diseases differ with respect to the mode of transmission (e.g., physical contact, air borne etc.), the causal pathogen, the incubation period etc. Due to this, there are differences in proposed approaches for identifying patterns and possible trajectories for the spread of the different infectious diseases. Research in the field of disease transmission focuses on inferring of transmission trees (outbreak - specific), parameters (transmission model-dependent), and identifying patterns in pathogen mutation of diseases in a given area based on epidemiological data and gene sequencing. Before the introduction of sequenced genome in the reconstruction of transmission networks, contact tracing was the main avenue used in the building trees to identify spread patterns (Stimson et al. 2019). This process is however tedious, cost intensive and results in less precise transmission trees. The introduction of genome sequencing has been a major breakthrough in this field, and epidemiological data (i.e. infection times, number of infected cases, distribution of prevalence, incidence rate, etc.) combined with genome data helps in inferring more accurate transmission trees for outbreaks (Worby et al. 2016).

A number of transmission models have been developed to analyze the spread of Ebola virus (Kraemer et al. 2019), H7N7 avian Influenza (Hall et al. 2015), Foot and mouth disease (Cottam et al. 2006, 2008; Morelli et al. 2012), Tuberculosis (Didelot et al. 2014; Guerra-Assuncao et al. 2015) among others. These methods make use of genetic data and consider either Bayesian or frequentist models in reconstructing transmission trees for the epidemic under study.

Kenah et al. (2016) categorize the approaches to reconstruct transmission trees and estimate parameters of transmission models based on how epidemiological data is combined with genetic data in the reconstruction of transmission trees. The first approach is where transmission trees are reconstructed using genetic distances from the genome sequenced from infected cases. Wertheim et al. (2011) construct a transmission network by linking cases based on the number SNP differences. Similarly, Guerra-Assuncao et al. (2015) also reconstruct transmission networks for the spread of Tuberculosis in a district in Malawi using the number of SNP differences as the basis for inferring connections between cases.

The second approach uses weights of possible infector-infectee connections obtained from pseudolike-

lihood models based on genetic and epidemiological data. By finding the product of components based on epidemiological and genetic data, the weight of each possible link in the tree is obtained, with the joint product of the edge weights being the weight of the complete transmission. Morelli et al. (2012), Ypma et al. (2012), use this approach in the reconstruction of transmission trees by incorporating spatial, epidemiological and genetic data in the computation of link weights.

The third approach is where a phylogenetic tree is obtained from sequenced genome and combined with epidemiological data in reconstructing the transmission trees. Cottam et al. (2008), Gardy et al. (2011) use this approach in the reconstruction of transmission trees for the foot-and-mouth disease virus outbreak among farms in the UK and a tuberculosis outbreak in British Columbia respectively. This approach has been undergoing constant refining and approaches have been developed to simultaneously infer both the phylogenetic tree and the transmission tree using Markov Chain Monte Carlo (MCMC) methods (Ypma et al. 2013; Didelot et al. 2014).

Firestone et al. (2019) compared 12 different approaches used in reconstructing transmission trees by assessing their ability to accurately reconstruct the underlying transmission network. Even though the results indicated that some approaches performed better compared to others, they state that a framework for reconstructing a transmission tree may suit a particular disease but may not necessarily be the best approach for reconstructing transmission networks of another disease given the disease dynamics and epidemiological parameters.

1.1 Modelling Tuberculosis transmission in British Columbia

Transmission trees help in identifying the main source of an outbreak as well as the pattern of the spread of the disease. This information is very important in the control and curtailing of the disease. Information obtained from infected individuals help identify their contact pattern and based on the available epidemiological data, we can infer other possible transmissions. This aids in identifying areas that need quick responses in order to prevent an outbreak. With Tuberculosis (TB) being a problem in British Columbia (BC), we seek to develop models that can explain transmission patterns and aid in making decisions on interventions. The BC Centre for Disease Control (BCCDC) reports that there are approximately 250 cases of TB diagnosed each year and they serve as the main point of contact for all TB-related laboratory, epidemiological, clinical and public health activities. These records on infected patients can be used in reconstructing TB transmission trees in BC and provide health services and

stakeholders with information on how to distribute resources.

Didelot et al. (2014) study how within host-diversity of the pathogen that causes tuberculosis (Mycobacterium) affects the reconstruction of transmission trees in densely-sampled outbreaks. They show that incorporating this factor helps improve the transmission trees constructed. Didelot et al. (2017) extend this approach to consider the reconstruction of transmission trees in the case of partially sampled and ongoing outbreaks.

However, their approach focuses on the reconstruction of a single transmission tree, using a single phylogenetic tree obtained from the sequenced genome. Kenah et al. (2016) argue that reconstructed transmission trees for an outbreak of a disease at a given time does not necessarily translate to the transmission tree for another outbreak of the same disease. However, these trees help in obtaining estimates of transmission parameters that are useful in identifying risk factors, as well as panning out health interventions and strategies to deal with present and future cases of an outbreak.

Kenah et al. (2016) propose the use of a survival analysis framework in modelling transmission transmissions. Here, the time to an infectious contact (i.e. a susceptible individual makes contact with an infectious person) is modelled and censoring is done with respect to an infectious contact interval. Kenah et al. (2016) show that parameter estimates of the hazard of infectious contact can be obtained by summing or averaging over all possible transmission trees. Kenah et al. (2016) also show that phylogenetic trees can be incorporated into this framework to reduce the number of possible trees obtained.

There is, however, uncertainty associated with the phylogenetic tree used in estimating transmission parameters. For a set of sequenced genome, the given phylogenetic tree could take several forms. Stochastic models which describe the demographics of the outbreak through time (e.g. the number of infections, growth rate of infections, death rate etc.), the evolution of the pathogen during outbreak and the distribution of parameters, are used in a Bayesian framework to obtain the posterior distribution of the phylogenetic trees. However, using these approaches and selecting a single phylogenetic tree when inferring transmission trees and estimating model parameters ignores the fact that the chosen phylogenetic tree is not "the" true phylogenetic tree.

With the availability of epidemiological, genetic and spatial data of TB in British Columbia, we propose a survival analysis framework that adequately accounts for variability in phylogenetic trees in estimating the transmission parameters. We also seek to assess the predictive power of our proposed

approach.

1.2 Research Questions

The research questions of interest are listed as follows:

1. What is a valid approach to account for the uncertainty in generated phylogenetic trees in the reconstruction of transmission trees using a survival analysis framework?

2. Transmission models are often modified to include covariates that account for different dynamics in the population. The inclusion of these components may improve the performance of the transmission models as shown by Firestone et al.(2019) when they included a spatial component in the model proposed by Cottam et al.(2008). With the availability of spatial data on infected TB cases in British Columbia, what is a valid approach to incorporate spatial-temporal components into the survival analysis framework for transmission tree reconstruction?

3. What are appropriate validation methods to assess prediction improvement in transmission trees and the estimation of transmission parameters when incorporating multiple phylogenetic trees and spatial-temporal components.

This proposal presents a method for the first research question. We use data from the "Darlington" cluster during the Foot and Mouth Disease (FMD) outbreak that occurred in the UK in 2001 which was used by Kenah et al. (2016) in assessing the performance of their proposed method.

The remainder of this proposal is as follows. Section 2 is a literature review of proposed approaches for the inference of transmission trees using genetic, epidemiological and spatial data. Section 3 describes the framework for inference for Phylogenetic trees, Section 4 describes the framework for Susceptible-Exposed-Infectious-Removed (SEIR) model and how it is combined with survival analysis in modelling outbreaks. In Section 5, we show how variants of the Expectation-Maximization algorithm can be used in estimation of the parameters of the hazard transmission model accounting for the uncertainty in the generate phylogenetic trees, and provided an overview of the implementation of our proposed approach in Section 6. We present results of applying the proposed approach to the FMD outbreak in the UK and simulations in Section 7 and discuss the further work to be done with regards to building a theoretical framework for accounting for uncertainty of the phylogenetic trees as well as incorporating spatial com-

ponents and assessing models that can be used to predict or identify potentially undiagnosed TB cases from healthcare contact data in Section 8.

2 Literature Review

2.1 Reconstructing transmission trees

Many approaches have been proposed in the literature for the reconstruction of transmission trees for infectious diseases. In recent years, there has been progress in the development of frameworks that incorporate genetic data and epidemiological data in inferring transmission trees of outbreaks (Jombart et al. 2014). Cottam et al. (2008) combine genetic and epidemiological data to reconstruct a transmission tree for the 2001 FMD outbreak among farms in the UK. They show that the incorporation of epidemiological data that relate to the timing of infection and infectiousness, as well as the genetic data, produces trees that are closer to the true transmission pattern. Ypma et al. (2012), Morelli et al. (2012) and Jombart et al. (2014) also propose approaches that combine epidemiological data and genetic data in the reconstruction of the transmission pattern for the epidemic of avian influenza A (H7N7) in the Netherlands in 2003, UK epidemics of foot-and-mouth disease virus in 2007, and the 2003 Singaporean outbreak of Severe Acute Respiratory Syndrome (SARS), respectively. Didelot et al. (2014) account for the within-host genetic diversity of the mycobacterium tuberculosis pathogen by modelling its evolution as a neutral coalescent process with a constant population size and a fixed duration for the replication cycle before using it in the reconstruction of transmission trees of a tuberculosis outbreak.

2.1.1 Adding phylogenetic trees

Existing approaches where phylogenetic trees are incorporated in the reconstruction of the transmission trees have either employed a two-step procedure for the inference of transmission trees or a joint estimation procedure for the phylogenetic and transmission trees. For the two-stage procedure, a single phylogenetic tree is obtained based on the sampled genetic data. Using this structure along with the epidemiological data, a frequentist or Bayesian approach is employed to estimate the transmission tree for the given outbreak. Cottam et al. (2008) apply this approach in reconstruction of transmission trees for the 2001 foot-and-mouth disease outbreak in the UK. They obtained gene sequences from animals in infected farms and constructed a phylogenetic tree using the TCS software package developed by Clement et al.

(2000) which establishes genealogical relationships based on statistical parsimony. Cottam et al. (2008) use a recursive algorithm which works backwards from the tip of the phylogenetic tree, identifying the most recent common ancestor of a pair of cases, to generate possible transmission trees. They use a maximum likelihood approach to estimate the likelihood of the transmissions trees and the parameters of the transmission model. The knowledge of transmission routes obtained from the phylogenetic tree significantly reduces the possible transmission trees inferred from the observed data. This approach is employed by Kenah et al. (2016) in developing an algorithm that reduces the list of possible transmission trees by identifying trees which are consistent with a generated phylogenetic tree. Didelot et al. (2014) also apply the two-stage procedure in inferring transmission trees for a densely-sampled tuberculosis outbreak while accounting for the within-host genetic diversity (i.e., the mutation of the pathogen within each host after they are infected). They first obtain a time-labeled phylogenetic tree using the Bayesian Evolutionary Analysis Sampling Trees (BEAST) software developed by Drummond et al. (2002), and then infer a transmission tree. Using a simulation study, they show that the inclusion of the within-host genetic diversity results in more accurate transmission trees. Didelot et al. (2017) employ this approach in inferring transmission trees for partially sampled and ongoing outbreaks. They developed the *Transphylo* package in *R* for the implementation of the proposed approach.

Methods that simultaneously estimate phylogenetic and transmission trees account for the fact that lineages in a single phylogenetic tree may not be compatible with the observed epidemiological data. They seek to explore the space for both transmission and the phylogenetic trees. Hall et al. (2015) argue that a fixed phylogenetic tree restricts the space of transmission trees, and some transmissions may be wrongly assigned in the final transmission tree. They develop an MCMC procedure to jointly reconstruct transmission and phylogenetic trees allowing for the exploration of the entire space of transmission and phylogenetic trees. They apply the proposed approach to the H7N7 avian influenza outbreak in the Netherlands in 2003. Their work builds on the methods developed by Didelot et al. (2014) and helps estimate more precise phylogenetic trees by making use of information from the epidemiological data. The proposed approach is incorporated in the BEAST software using the *BeastLier* package. Klinkenberg et al. (2017) also propose a Bayesian framework which combines models for transmission, within-host genetic dynamics, and mutation to simultaneously infer the transmission and phylogenetic trees given the genetic data and the sampling times. To account for possible intra-host evolution of the pathogen, they use a

hierarchical tree perspective in which a phylogenetic tree is described as a collection of mini-trees which are connected through transmission events. This reflects the pathogen evolution within hosts. Klinkenberg et al. (2017) apply the proposed model to data sets on outbreak of Tuberculosis (Didelot et al. 2014), Methicillin-resistant Staphylococcus aureus (Nubel et al. 2013), foot-and-mouth disease (Cottam et al. 2008, Morelli et al. 2012, Ypma et al. 2013) and the H7N7 avian influenza (Ypma et al. 2013; Hall et al. 2015). They obtain similar or improved results compared to previously implemented methods for reconstructing transmission trees for these outbreaks. The proposed method is implemented in R using the *Phybreak* package.

For methods where the phylogenetic trees are used in inferring transmission trees, a number of approaches exist for summarizing trees or finding a consensus phylogenetic tree from a sample of trees typically obtained using Markov Chain Monte Carlo methods. Bryant (2003) reviews methods used in phylogenetics, and identifies links between the different methods. However, these approaches may not necessarily choose a good representation of the phylogeny. Drummond et al. (2002) implement a number of different approaches in summarizing the phylogenetic trees in the BEAST software. A majority rule consensus tree is constructed by including all clades that occur in at least 50% of trees in the posterior distribution. The maximum clade credibility (MCC) tree takes the tree with the maximum product of the posterior clade probabilities. The MCC tree is always one tree chosen from the set of trees, and is often shown in publications that use BEAST for phylogenetic reconstruction (Drummond and Bouckaert, 2015). The maximum a posteriori (MAP) tree topology is the tree topology that has the greatest posterior probability, averaged over all branch lengths and substitution parameter values. Another approach is to define the median tree which is quite a difficult task. This approach entails visualizing the trees in the posterior sample as a cluster of points in a high-dimensional space. The tree in the middle of the cluster is the median tree. A metric that may be used to obtain the median tree is the Robinson-Foulds distance (Robinson-Foulds distance, 1981). Here, phylogenetic trees a represented a vector of vertices and edges, with the metric calculating the distance between two trees by finding the average number of unique partitions for both trees. The median tree has the shortest average distance to other trees in the posterior distribution.

The transmission and phylogenetic tree reconstruction methods have mainly focused on finding transmission trees that are close to representing the pattern of spread of a disease in an outbreak. However,

suggestions have been made to focus on the estimation of population parameters that relate to the transmission of diseases. Kenah et al. (2016) argue that the trees reconstructed for a given outbreak may not translate to future outbreaks. They propose that inferred transmission trees be used in the estimation of transmission model parameters which can help in identification of risk factors and planning of interventions.

2.1.2 Survival Analysis in disease transmission

The use of survival analysis in modelling transmission of diseases is a growing field. Lau et al. (2015) proposed a Bayesian framework for integrating genetic and epidemiological data using survival analysis based likelihoods. Kenah et al. (2016) implement the two-stage procedure for the reconstruction of transmission trees and the estimation of transmission model parameters based on survival analysis models. They use a single phylogenetic tree and developed an algorithm that lists the set of possible transmission trees that are consistent with the phylogenetic tree and epidemiological data. However, the use of a single phylogenetic tree ignores the uncertainty associated with choosing one phylogenetic tree out of the many possible trees and this may also affect the estimates of parameters of the models used in the reconstruction of the tree.

We build on the approach of Kenah et al. (2016) by accounting for all possible phylogenetic trees in the estimation of parameters. We treat the phylogenetic trees as unobserved data and use a variation of the EM algorithm in the estimation of model parameters.

2.1.3 Spatial data in disease transmission

Incorporating spatial data into the framework of reconstructing transmission trees for infectious diseases is also being increasingly recognized as an important component in reconstructing transmission trees (Riley et al. 2014). In a review of large-scale spatial transmission models, Riley (2007) considers the importance of households in studies on disease transmissions. The paper identifies that in a new outbreak or re-emerging pathogen, disease incidence tend to occur in spatial clusters and spatial models are implemented to help analyze the pattern of spread of the disease and also assess the impact of interventions. For location data (i.e., data on the geographical location of the home of individuals or herd of animals), transmission models which use the distance between two fixed locations can be employed. Keeling et al. (2001) study the dynamics of the 2001 foot-and-mouth outbreak in UK where they set the farms as the

individual units of infection. They develop an individual farm-based stochastic model and show that the spatial distribution, size, and species composition of farms all influence the observed pattern and regional variability of outbreaks. In particular, they identified areas that were potential "hot spots" for infection and observed that the introduction of movement restrictions between farms highly skewed the spatial infection distribution.

Some recent approaches have integrated spatial data along with epidemiological data and genetic data in the reconstruction of transmission trees. The distance between the locations of an infectious case and a susceptible case affects the likelihood of an infection occurring. Aldrin et al. (2011) present a stochastic spatio-temporal model for the rate of infectious salmon anaemia among Norwegian salmon farms. Taking into account the seaway distance (i.e., the shortest route between two points via water) and genetic distances, they identified the few farms that were the most probably sources of infections and this helped in mapping the infection pathways. Morelli et al. (2012) proposed a Bayesian Inference framework to reconstruct transmission trees where they include spatial data as an observed covariate in estimating the posterior distribution of transmission trees. They assume that the spatial location of premises affects the infection potential between any two premises and the probability of a premise j infecting a premise i decreases exponentially with the absolute distance between the two locations. The proposed framework is applied to the 2001 and the 2007 UK epidemics of foot-and-mouth disease virus. The model robustly infers transmission pathways, and uncovers the presence of undetected farms in the transmission network. Mollentze et al. (2014) also describe a statistical framework for inferring partially observed transmission events and estimating the number of unobserved infections. Using a generalization of the proposed model by Morelli et al. (2012), they modify the spatial component to accommodate a wide variety of spatial transmission patters by replacing the exponential transmission kernel used in Morelli et al. (2012) with an exponential-power spatial transmission kernel. These transmission kernels assume that the correlation structure existing between two locations is known in modelling the distance between the cases (Diggle and Ribeiro Jr. 2007).

Lau et al. (2015) incorporate spatial data in their SEIR epidemiological model by characterizing the rate of infection between an infector and infectee with a spatial kernel function which incorporates the distance between the two cases. They assume an exponential kernel which indicates that the rate of infection decreases with distance. Hall et al. (2015) use three models to describe the epidemiological

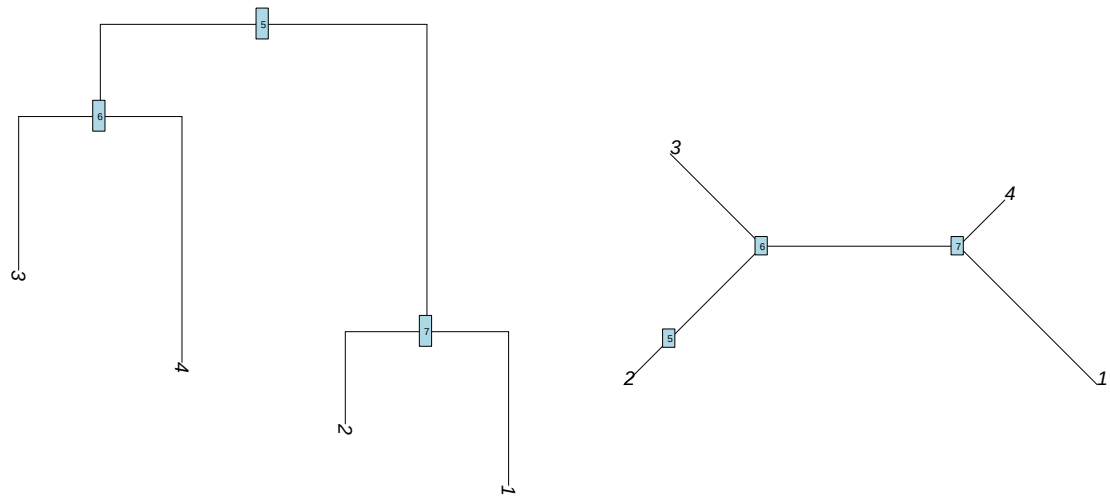
and evolutionary process involved in an outbreak: a stochastic model of infection and between-host transmissions, a deterministic model of the within-host pathogen evolution, and a stochastic model of sequence evolution. Amongst these three, the spatial component is incorporated into the stochastic model that models the infection and between-host transmission dynamics. An infectious case is assumed to inflict constant force of infection on a susceptible case at a given rate and this rate is modified by a function which accounts for the geographical distance between the infectious case and the susceptible case. They apply the proposed approach to the 2003 H7N7 avian influenza outbreak that occurred in the Netherlands and show that the model performs well for the reconstruction of transmission trees.

To assess the performance of proposed methods for reconstructing transmission trees for infectious diseases, Firestone et al. (2019) compared the performance of twelve networks models in reconstructing the 2001 FMD outbreak in the UK. The Bayesian method developed by Lau et al. (2015) was found to be the most accurate for inferring the transmission network and exposure times. An interesting finding from their analysis was the improvement in the performance of the frequentist approach by Cottam et al. (2008) when modified to incorporate spatial and contact-tracing data. The accuracy of inferring the transmission network increased from 49% for the original method by Cottam et al. (2008) to 67% for the modified version by Firestone et al. (2019). The modified model was only outperformed by the Bayesian model by Lau et al. (2015) which uses survival analysis models.

Including spatial data in the approach by Kenah et al. (2016) may also improve the precision of the estimates of parameters of the hazard of transmission function and the inferred trees. We consider this approach in reconstructing transmission trees and estimating parameters of the hazard of transmission of tuberculosis in British Columbia using the available health data.

3 Phylogenetic Trees

Phylogenetic trees are used to represent the genealogical relationship among species, among genes, among populations, or even among individuals (Yang, 2006). These trees are connected graphs of nodes (vertex) and branches (edges) which describe the evolutionary process. The tips of the trees, which represent the sampled cases at a given time point, are known as leaves or external nodes. The nodes of unsampled cases observed in some time past are referred to as internal nodes. Trees can either be rooted or unrooted. A rooted phylogenetic tree has one internal node identified as its root and the start of the evolutionary



(a) Rooted (b) Unrooted

Figure 1: Rooted and unrooted phylogenetic trees

process. Unrooted trees on the other hand do not have a unique root indicating that the starting point of the tree is unknown. A special form of the rooted phylogenetic tree is a timed-tree which has branch lengths corresponding to the duration between branching events. For a timed-tree, the times of the internal nodes correspond to divergence or coalescent times, while the times of the leaves correspond to sampling times.

Trees can be classified as bifurcating or multifurcating based on the number of children for each internal node. For bifurcating (binary) trees, each internal node has two children whereas multifurcating trees have more than two children for each internal node. A rooted binary tree with n leaves has $2n-1$ nodes (internal and external) and $2n-2$ branches, while an unrooted tree with n leaves has $2n-2$ nodes and $2n-3$ branches. Figure 1 displays a rooted (Figure 1a) and unrooted (Figure 1b) phylogenetic tree with four leaves.

3.1 Inferring Phylogenetic trees

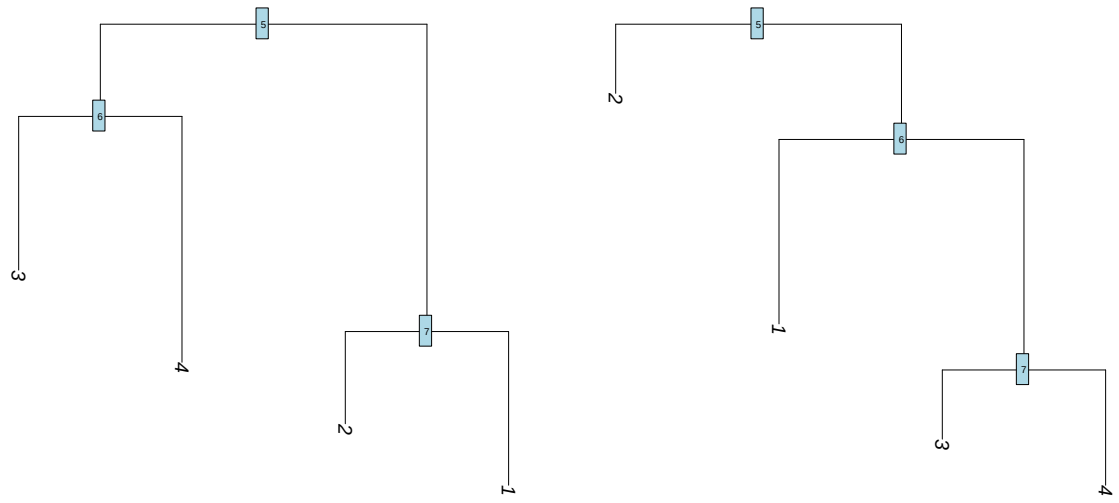
3.1.1 Number of trees and branch lengths of rooted trees

For any given number of sampled cases, there is a finite set of possible trees that can be generated. However, the list of possible trees increases exponentially as the number of sampled cases increases and this makes it close to impossible to enumerate and work with all possible phylogenetic trees with a moderate number of leaves. Focusing on binary rooted phylogenetic trees, the number of possible binary

phylogenetic trees for n sampled cases is (Cavalli-Sforza and Edwards 1967):

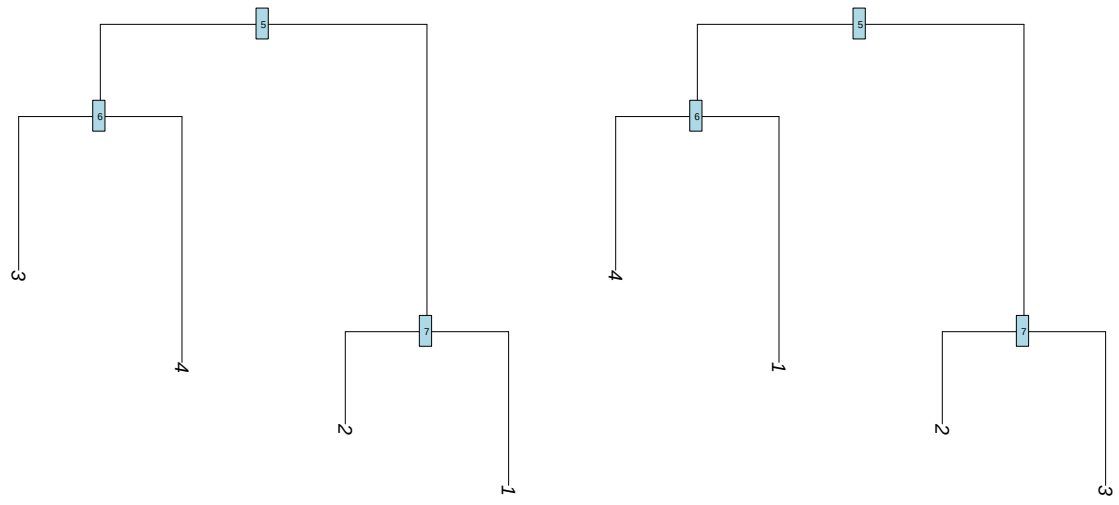
$$\prod_{k=2}^{n}(2k-3) = \frac{(2n-3)!}{2^{n-2}(n-2)!}, \quad n \geq 2\,.$$

Trees here differ in relation to the placement of the cases at the tips of the tree and inherently by the connection of sub-trees. The topology (structure) of two phylogenetic trees may be different based on the connections or placement of nodes in the trees. Two trees may have the same connections among sampled cases (representing the transmission pattern) but a different phylogenetic structure whereas another set of trees may have the same topology but may have different connections (transmission pattern) among sampled cases based on their placement at the tips of trees. For example, Figure 2 shows two phylogenetic trees with different topologies, however these two have the same connections between cases for the case where sampled case 2 is the root case which infects case 1 and case 3, and case 3 infects case 4. On the other hand, the two phylogenetic trees in Figure 3 have the same topology but different connections. We can have case 2 infecting case 3, while case 3 infects case 4 in Figure 3a but this connection is not possible in Figure 3b due to the placement of the leaves at the tips.



(a) tree 1 (b) tree 2

Figure 2: Different topology for phylogenetic trees but same connections



(a) tree 1 (b) tree 2

Figure 3: Same topology for phylogenetic trees but different connections

Branch lengths represent the number of mutations or state changes of measured traits that occur on a given branch (Felsenstein, 2004). These traits may include continuous character traits, e.g. height, number of legs and discrete (DNA, Amino acids) character traits, e.g. the number of DNA state changes along a branch for sequenced data from a sample of HIV infected persons. The mutation of sequenced gene data is modelled as a Markov process and we review models used in practice in section 3.2.1.

A number of approaches have been developed for inferring phylogenetic trees. These approaches seek to find the tree that best describes the data (e.g. DNA sequences) obtained from the sampled cases. These methods for inferring phylogenetic trees can be classified under three main headings namely: parsimony, distance, and likelihood-based methods. We briefly describe the first two categories in the next section and provide a detailed breakdown of the likelihood-based methods in subsequent sections.

3.1.2 Parsimony and Distance Methods

Parsimony and distance based methods make use of differences and distances between pairs of sampled cases. Focusing on sequenced DNA, parsimony methods seek to find the simplest or "most parsimonious" tree that best describes the observed data by finding the tree with the lowest number of evolutionary changes. Evolutionary changes here refer to the number of state changes of the DNA that need to occur to obtain the observed sequence given a particular tree. Due to the fact that the number of possible trees for a given number of cases grows exponentially, computing the number of evolutionary changes for all

trees may be an impossible task. Search algorithms that scan through the space of trees to find the best tree have been developed to help quicken this process. The nearest-neighbour interchange, is an algorithm where nearest neighbours in a tree are swapped before the number of changes is recorded. If the new tree has a lower score, it serves as the base tree and the swapping process is re-applied until a possibly optimal tree is obtained. Another approach is the *subtree pruning and regrafting* where branches are cut from one end and placed at another point before the number of changes is obtained. A limitation to these approaches is that they can get stuck in a local optimum.

For distance methods, the number of dissimilarities that exists between each pair of organisms in the data is obtained to form a distance matrix. The idea is that, cases that are close to each other and share a more recent common ancestor should have very few differences between the observed data. The best tree is then obtained by using algorithms such as the unweighted pair group method with arithmetic mean (UPGMA) and the neighbor joining algorithms. The UPGMA algorithm constructs the best tree by finding the shortest distance in the matrix, collapsing the two cases as one, reconstructing the distance matrix and repeating the process. The process is stopped when there are only two clusters left in the distance matrix. A limitation to the UPGMA is the fact that the rate of evolution on branches are assumed to be constant on all branches but this may not be the case. An alternative method which accounts for the unequal evolutionary rate is the neighbour joining algorithm which works similar to the UPGMA. A detailed breakdown of these approaches in provided by Felsenstein (2004).

3.1.3 Likelihood-based methods

The distance and parsimony methods select trees based on a given criterion but do not provide a framework to perform inference or/and express sample variability. Likelihood and Bayesian approaches provide a probabilistic framework for inferring phylogenetic trees. The likelihood approach computes the likelihood of the trees and picks the tree with the highest likelihood while the Bayesian approach on the other hand seeks to obtain the posterior probability of trees given the observed data set. Here, priors are assigned to parameters and Markov Chain Monte Carlo simulations are used to generate samples from the posterior distribution. We provide a breakdown of the two approaches in the next two sections.

3.2 Likelihood method for phylogenetic trees

Throughout this proposal, we describe the likelihood and Bayesian approaches based on sequenced DNA, which is a discrete character data set. In the likelihood framework, the tree with the highest likelihood is chosen as the best tree. Just as in the case of maximum likelihood estimation where we want to estimate the unknown parameters in a probabilistic data generating model, the unknown tree topology and branch lengths are the parameters of interest. A key component of likelihood based methods is the use of probabilistic models for DNA evolution. These models are used to calculate the probability of state changes along the branches of a tree. We briefly describe the substitution models before returning to the estimation of the likelihood of a tree.

3.2.1 Substitution models

The distance or difference between two sequences is defined as the expected number of nucleotide substitutions per site (Yang, 2006). The rate of evolution may be constant or vary overtime. In phylogenetic reconstruction, a probabilistic model is used to describe these changes between nucleotides. A continuous-time Markov process is commonly used in describing these evolutionary changes. DNA substitution models are specified over a four letter alphabet $\mathbf{U} = \{A, C, G, T\}$ with 12 transition rates in an instantaneous rate matrix (Drummond and Bouckaert, 2015).

Let q_{ij} denote the instantaneous rate of substitution from nucleotide i to nucleotide j with $i, j \in \{A, C, G, T\}$. Thus the *instantaneous transition rate matrix* is

$$Q = \begin{bmatrix} \cdot & q_{AC} & q_{AG} & q_{AT} \\ q_{CA} & \cdot & q_{CG} & q_{CT} \\ q_{GA} & q_{GC} & \cdot & q_{GT} \\ q_{TA} & q_{TC} & q_{TG} & \cdot \end{bmatrix}, \tag{1}$$

with the diagonal entries $q_{ii} = -\sum_{j \neq i} q_{ij}$, so that the rates for a given state should sum to zero across the row. The off-diagonal entries are positive and represent rates of flow of nucleotide i to nucleotide j.

At equilibrium, the expected mutation rate for each site i per unit time is:

$$\mu = -\sum_i \pi_i q_{ii}, \tag{2}$$

where π_i is the probability of being in state i at equilibrium, so that μ is just the weighted average outflow rate at equilibrium.

Let $W(t)$ represent the state of a Markov process W at time t. Assuming that W is in state $i \in \{A, C, G, T\}$ at time t, then in the next infinitesimal moment Δt, the probability that the process transitions to state $j \in \{A, C, G, T\}$ is $q_{ij}\Delta t$. In symbols:

$$Pr[W(t + \Delta t) = j | W(t) = i] = q_{ij}\Delta t, \quad j \neq i\,.$$

In general, the transition probability matrix, $P(t)$, which provides the probability of being in state j after time t, given that the process began at state i is given as

$$P(t) = \exp(Qt)\,. \tag{3}$$

An example of a model used in computing the rate matrix and the resulting transition probabilities is the Jukes-Cantor model (Jukes and Cantor, 1969). For this process, all transitions have equal rates, and all bases have equal frequencies (i.e., $\pi_A = \pi_C = \pi_G = \pi_T = 1/4$). A Q matrix for the Jukes-Cantor model with an expected mutation rate of 1 per unit time is given as:

$$Q = \begin{bmatrix} -3 & 1 & 1 & 1 \\ 1 & -3 & 1 & 1 \\ 1 & 1 & -3 & 1 \\ 1 & 1 & 1 & -3 \end{bmatrix}. \tag{4}$$

The entries of the transition probability matrix for the Jukes-Cantor process are computed as follows:

$$p_{ij}(t) = \begin{cases} \frac{1}{4} + \frac{3}{4}\exp\big(-4t\big) & \text{if} \quad i = j \\ \frac{1}{4} - \frac{1}{4}\exp\big(-4t\big) & \text{if} \quad i \neq j \end{cases}, \tag{5}$$

where t is the evolutionary time in units of substitutions per site (or genetic distances).

The F81 model (Felsenstein 1981) allows for unequal base frequencies ($\pi_A \neq \pi_C \neq \pi_G \neq \pi_T$). The Q matrix is:

$$Q = \eta \begin{bmatrix} \pi_A - 1 & \pi_C & \pi_G & \pi_T \\ \pi_A & \pi_C - 1 & \pi_G & \pi_T \\ \pi_A & \pi_C & \pi_G - 1 & \pi_T \\ \pi_A & \pi_C & \pi_G & \pi_T - 1 \end{bmatrix}. \tag{6}$$

Q is normalized by setting $\eta = \frac{1}{1-\sum_i \pi_i^2}$. This model has three free parameters, since the constraint that $\pi_A + \pi_C + \pi_G + \pi_T = 1$ removes one degree of freedom (Drummond and Bouckaert, 2015). The transition probabilities are:

$$p_{ij}(t) = \begin{cases} \pi_i + (1 - \pi_j)e^{-\eta t} & \text{if} \quad i = j \\ \pi_j(1 - e^{-\eta t}) & \text{if} \quad i \neq j \end{cases}. \tag{7}$$

Other models used in the computation of the rate matrix and the resulting transition probabilities include the K80 model (Kimura, 1980), the HKY process (Hasegawa et al. 1985) and the general time-reversible (GTR) model which are described by Drummond and Bouckaert (2015) and Felsenstein (2004).

3.2.2 Likelihood of a phylogeny

Likelihood methods for phylogenetic trees estimate parameters (e.g., branch lengths ($\boldsymbol{\ell} = \{\ell_1, \ell_2, ...\ell_{2n-2}\}$) and evolutionary parameters $\boldsymbol{\kappa}$) by maximizing the likelihood, often using numerical algorithms (Yang, 2006). The data consists of n aligned sequences (sampled cases), each m sequences long, and can be represented as an $n \times m$ matrix $\mathbf{D}$, where d_{ih} is the hth nucleotide (site) in the ith sequence. For this approach, a nucleotide-substitution model is assumed for the data. Two crucial assumptions for this approach are:

1. Evolution at different sites (m) are independent (Yang, 2006).

2. Conditional on the internal node states, evolution proceeds independently on different branches of

the phylogeny.

Let $L(\Phi, \boldsymbol{\ell}, \boldsymbol{\kappa})$ be the likelihood corresponding to the $n \times m$ DNA sequence alignment matrix $\mathbf{D}$ for a given tree topology Φ with branch length vector $\boldsymbol{\ell}$ and the evolutionary parameters $\boldsymbol{\kappa}$. The likelihood is written as:

$$L(\Phi, \boldsymbol{\ell}, \boldsymbol{\kappa}|\mathbf{D}) = Pr(\mathbf{D}; \Phi, \boldsymbol{\ell}, \boldsymbol{\kappa}) = \prod_{h=1}^{m} Pr(\mathbf{D}_h; \Phi, \boldsymbol{\ell}, \boldsymbol{\kappa}) = \prod_{h=1}^{m} L_h(\Phi, \boldsymbol{\ell}, \boldsymbol{\kappa}) \tag{8}$$

where $\mathbf{D}_h$ is the $n \times 1$ vector of observed nucleotide at the hth site and $L_h(\Phi, \boldsymbol{\ell}, \boldsymbol{\kappa})$ is the site h likelihood. We compute the likelihood of the entire sequence using Felsenstein's pruning algorithm (1973) which involves finding the product of the likelihood of each site based on the assumption of independence. For a tree with three sampled cases as in Figure 4, where the observed DNA characters for a chosen site h are A,C and G for cases I, II and II respectively, the probability of the data at site h given the tree topology, branch lengths and substitution parameters is given as:

$$Pr(\mathbf{D}_h; \Phi, \boldsymbol{\ell}, \boldsymbol{\kappa}) = \sum_{x} \sum_{y} Pr(A, C, G, x, y; \ell, \Phi, \kappa) \tag{9}$$

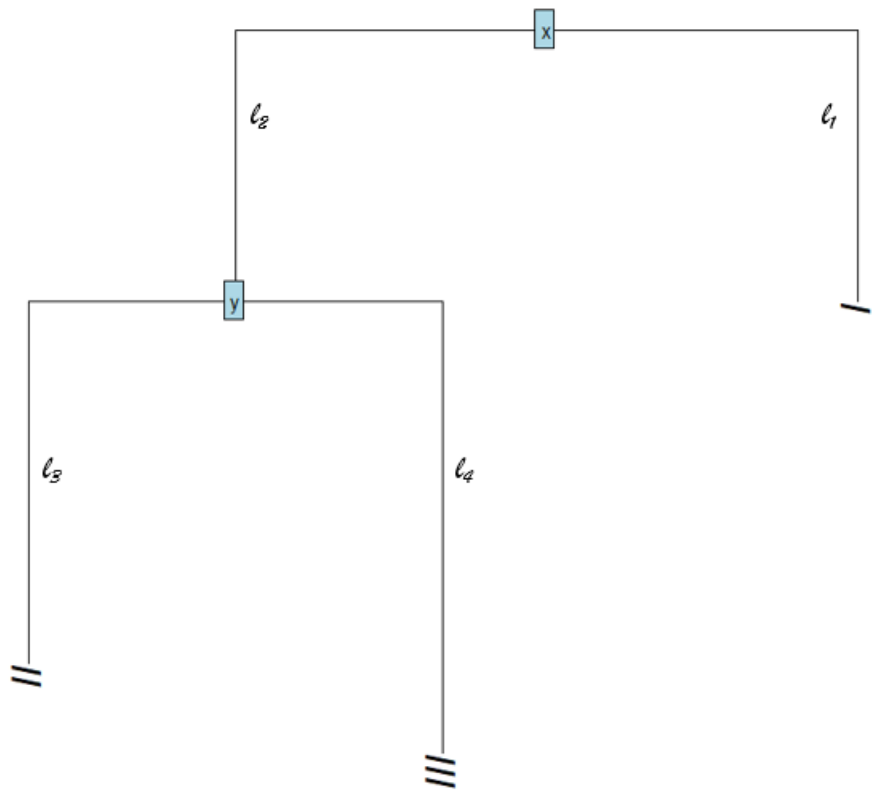


Figure 4: Example of Phylogenetic tree for 3 sampled cases

where x and y in Figure 4 are the character states of the internal nodes for site h. The summation is done over all four (A,C,G,T) DNA states. With the second assumption of independence of evolution among different lineages, the summation can be re-written into product terms such as

$$Pr(A, C, G, x, y; \boldsymbol{\ell}, \Phi, \boldsymbol{\kappa}) = \sum_x \sum_y Pr(x)Pr(y|x, \ell)Pr(A|y, \ell)Pr(B|y, \ell)Pr(C|x, \ell). \tag{10}$$

The expression in eq (10) helps in the computation of the likelihood.

This approach maximizes the likelihood with respect to the tree topology, the branch lengths and the substitution parameters. This requires a two-step maximization algorithm where optimal branch lengths and substitution parameters are obtained for a given tree topology in stage one, and then the optimal tree topology is obtained for the given branch lengths and substitution parameter. This, however, is a non-convex optimization and is solved in practice using search algorithms such as Newton-Raphson and Expectation-Maximization. Felsenstien (2004) describes the methods employed in the maximization of

the phylogenetic likelihood.

3.3 Bayesian methods for phylogenetic trees

For the Bayesian approach, the posterior distribution of the phylogenetic tree given the observed sequenced data is defined as

$$P(\Phi, \boldsymbol{\ell}, \boldsymbol{\kappa}|\mathbf{D}) \propto P(\mathbf{D}|\Phi, \boldsymbol{\ell}, \boldsymbol{\kappa})P(\Phi, \boldsymbol{\ell}|\boldsymbol{\kappa})P(\boldsymbol{\kappa}), \tag{11}$$

where $\mathbf{D}$ is the sequenced genomic data and, as before, Φ is the phylogenetic structure, $\boldsymbol{\ell}$ is the branch lengths, and $\boldsymbol{\kappa}$ is the collection of the evolutionary parameters, which includes parameters of the models describing the between-host and the within-host transmission dynamics. Models that describe the between-host and the within-host transmission dynamics are called population dynamics models and site models, respectively (Drummond et al. 2002).

3.3.1 Population dynamics model

The population dynamic model determines the probability $P(\Phi, \boldsymbol{\ell}|\boldsymbol{\kappa})$ taking into account how the population of infections changes over time. Most commonly used models are the coalescent and birth-death models. The coalescent model looks at how far back in time one has to go to find the most recent common ancestor (MRCA) for any two individuals. This approach relies on a collection of mathematical models to describe the population dynamics and parameterizes the phylogenetic tree in terms of the effective population size (N_e) and the mutation rate per generation time (ρ). The product of the two parameters is the mutation-scaled effective population size denoted by $\psi = N_e\rho$. This is a scale factor that converts "coalescent time" to calendar time and this is what is estimated in the model. The effective population size can either be assumed to be constant over time (Kingman, 1982) as presented in the constant-size population Wright-Fisher model or be assumed to grow at a certain rate over time. For example, in the Bayesian context, this rate may be described by an exponential distribution.

For a given phylogenetic tree Φ, with n tips and $n-1$ internal nodes and t representing the divergence times of the internal nodes, the probability density for a timed-tree Φ, generated from the coalescent

process, using the Wright-Fisher model is given as

$$P(\Phi|\psi) = \frac{1}{\psi^{n-1}} \prod_{i=2}^{2n-1} e^{(-k_i(k_i-1)/2\psi)(t_i-t_{i-1})}, \tag{12}$$

where k_i denotes the number of lineages present in the interval of time between node $i-1$ and node i, and $\psi = N_e\,\rho$ is a between-host evolutionary parameter included in the vector $\boldsymbol{\kappa}$ described in eq (11).

Non-parametric coalescence models that are flexible and allow for the estimation of complex demographics have also been developed. Examples include the Skyride and the Skygrid models. A detailed description of these models is provided in Drummond and Boukaert (2015).

The birth-death model assumes that through time, each individual gives rise to an additional offspring at a rate λ and dies with rate ξ over the period of the (Bouckaert et al. 2019). It is assumed that each speciation in the phylogenetic tree results in exactly two descendants. The process starts with one lineage at time t_0 in the past and continues forward in time until the present (time 0). An individual is sampled with probability ω. The probability density of a timed-tree produced after time (t_0) with n tips, given a birth rate λ, a death rate ξ, a sampling probability ω (Stadler, 2010) is given as

$$P(\Phi|\lambda,\xi,\omega,t_0) = p_1(t_0) \prod_{i=1}^{n-1} \lambda p_1(t_{n+i}), \tag{13}$$

with

$$p_1(z) = \frac{\omega(\lambda-\xi)^2 e^{-(\lambda-\xi)z}}{(\omega\lambda + (\lambda(1-\omega)-\xi)e^{-(\lambda-\xi)z})^2},$$

where $p_1(z)$ is the probability of an individual at time z in the past having precisely one sampled descendant, and and λ, ξ, and ω are between-host evolutionary parameters included in the vector $\boldsymbol{\kappa}$ described in eq (11). The model in (13) is known as the constant-rate birth-death model. Other forms of the birth-death model allow the birth and death rate parameters to change through time. Drummond and Bouckaert (2015) present examples of the other forms of the birth-death models.

3.3.2 Likelihood of observed sequence

To compute the likelihood of the sequenced data ($\mathbf{D}$), given the phylogenetic tree and substitution parameters [$\mathrm{P}(\mathbf{D}|\Phi,\boldsymbol{\ell},\boldsymbol{\kappa})$] in eq (11), let the sequenced genomic data of a tree with n leaves (sampled cases)

be represented by

$$\mathbf{D} = \begin{pmatrix} \mathbf{D}_{I_1} \\ \mathbf{D}_{I_2} \\ \vdots \\ \mathbf{D}_{I_n} \end{pmatrix} = \begin{pmatrix} s_{I_1,1} & s_{I_1,2} & \dots & s_{I_1,m} \\ s_{I_2,1} & s_{I_2,2} & \dots & s_{I_2,m} \\ \vdots & \vdots & \ddots & \vdots \\ s_{I_n,1} & s_{I_n,2} & \dots & s_{I_{n-1},m} \end{pmatrix} \tag{14}$$

where I_k is the index of the kth leaf and D_{I_k} is the sequence of length m associated with the leaf. The entry $s_{I_k,j}$ represents the nucleotide base at site j, and takes values in the set letter $\mathbf{U} = \{A, C, G, T\}$.

Let $\mathbf{D}_Y$ be an $(n-1) \times m$ matrix representing the unknown ancestral sequences of the internal nodes of the phylogenetic tree:

$$\mathbf{D}_Y = \begin{pmatrix} \mathbf{D}_{Y_1} \\ \mathbf{D}_{Y_2} \\ \vdots \\ \mathbf{D}_{Y_{n-1}} \end{pmatrix} = \begin{pmatrix} s_{Y_1,1} & s_{Y_1,2} & \dots & s_{Y_1,m} \\ s_{Y_2,1} & s_{Y_2,2} & \dots & s_{Y_2,m} \\ \vdots & \vdots & \ddots & \vdots \\ s_{Y_{n-1},1} & s_{Y_{n-1},2} & \dots & s_{Y_{n-1},m} \end{pmatrix} \tag{15}$$

where Y_r is the index of the rth internal node and D_{Y_r} is the sequence of length m associated with the internal. Let $\mathtt{L}$ denote the set of all possible ancestral sequences. For a given edge $\langle i, j \rangle$, belonging to set of edges of a tree Φ, an individual associated with node j is a direct descendant of the individual associated with node i. These two individuals may however have differences in their sequences $\mathbf{D}_i$ and $\mathbf{D}_j$ due to substitutions that have occurred in the interval t_{i-1} and t_i. Applying a continuous-time Markov process model of substitution gives $Pr(s_{j,k} = u^{'} | s_{i,k} = u) = \left[e^{Q\mu(t_i - t_j)}\right]_{u,u'}$. Then the probability for any particular set of sequences $\mathbf{D}$ to be realised at the leaf nodes of the tree is:

$$P(\mathbf{D}|\Phi, \boldsymbol{\ell}, \boldsymbol{\kappa}) = \sum_{\mathbf{D}_Y \in \mathtt{L}} \prod_{\langle i,j \rangle \in R_\Phi} \prod_{k=1}^{m} \left[e^{Q\mu(t_i - t_j)}\right]_{s_{i,k}, s_{j,k}} \tag{16}$$

where R_Φ represents the edge set (connections from a tip to an internal node) of Φ. Calculating the sum over all possible ancestral sequences D_Y in eq(16) is feasible using a pruning algorithm (Felsenstein, 1981).

With the development of models to describe the between-host and within-host dynamics, a search

through literature shows that there is not much work done on the joint estimation of transmission and phylogenetic tree parameters using the frequentist approach. This may be due to the fact that there is another level of complexity introduced when these phylogenetic models are combined with transmission tree models. In this proposal we consider an EM approach to jointly estimate the transmission and phylogenetic tree parameters.

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3.4 Software for Phylogenetic Trees

A number of software packages have been developed to implement the methods described here for the inference of phylogenetic trees. The BEAST software implements the Bayesian framework described in section 3.3. Other software that implement the Bayesian method include: MrBayes (Huelsenbeck and Ronquist, 2001), SeaView (Gouy et al., 2010) and PhyloBayes (Lartillot and Philippe, 2004; Lartillot and Philippe, 2006; Lartillot et al., 2007). Likelihood-based methods implementations include: PHYLIP (Felsenstien, 1989), PhyML (Guindon et al., 2010), PAML (Yang, 1997) and FastTree (Price et al., 2009).

4 SEIR and Survival Framework

We begin by considering a stochastic SEIR model which is used to model the evolution of an outbreak and determines the state of each individual in an infected area at a given time. This model does not provide a transmission tree but helps in identifying those at risk of infection, those capable of infecting others and those who have recovered from an infection at any given time point.

At every point in time an individual is considered to be in either of 4 states: susceptible (S), exposed (E), infectious (I) or removed (R). An infected individual i moves from S to E at time t_i, with $t_i = \infty$ if i is never infected. After infection, there is a latency period ε_i ($\varepsilon_i \geq 0$) after which the individual moves from state E to I and becomes infectious at time $t_i + \varepsilon_i$. The infectious period for the individual is denoted $l_i (l_i > 0)$ and during this period, the individual can infect other individuals. At time $t_i + \varepsilon_i + l_i$ the individual moves from state I to R and can no longer infect others or be infected. The latency period (ε_i) and infectious period (l_i) are random variables with finite mean and variance.

An infectious contact is a contact that is sufficient to infect a susceptible person. Given that individual i is infectious at time $t_i + \varepsilon_i$, an infectious contact is made with subject $j \neq i$ at time $t_{ij} = t_i + \varepsilon_i + \tau^*_{ij}$.

The infectious contact interval τ^*_{ij} can only be observed if individual j is infected by individual i. τ^*_{ij} will not be observed if either: (a) individual i recovered before making an infectious contact with j, (b) if individual j is infected by another individual before coming into contact with individual i, or (c) if individual j is no longer under observation when it came in contact with individual i. The infectious contact interval τ^*_{ij} is therefore censored, with $\tau^*_{ij} \in (0, l_i]$ if i infects j and $\tau^*_{ij} = \infty$ when i never infects j.

Let C_{ij} be an indicator variable which takes the values 1 if an infectious contact occurred between i and j, and 0 otherwise. Let $I_i(\tau) = \mathbf{1}_{\tau \in (0, l_i]}$ be a left-continuous process indicating whether i remains infectious at infectious age τ. Let $S_{ij}(\tau) = \mathbf{1}_{t_i + \varepsilon_i + \tau \leq t_j}$ be a left-continuous process indicating whether j remains susceptible when i reaches infectious age τ. We assume that the population is under observation until a stopping time T and $O_{ij}(\tau) = \mathbf{1}_{t_i + \varepsilon_i + \tau \leq T}$ be left-continuous process indicating whether j is under observation when i reaches infectious age τ. Then

$$Y_{ij}(\tau) = C_{ij} I_i(\tau) S_{ij}(\tau) O_{ij}(\tau) \tag{17}$$

is a left-continuous process indicating whether infectious contact from i to j can be observed at infectious age τ of i. The assumptions above ensure that censoring of τ_{ij} is independent for all ij.

4.1 Survival Analysis for transmission data

The infectious contact interval between individual i and individual j, $\mathcal{T}_{ij}$, is a random variable assumed to be drawn from a distribution with hazard function $h_{ij}(\tau)$. Let the hazard of infectious contact from i to j at time τ after the onset of infectiousness in i be

$$h_{ij}(\tau) = \frac{f_{ij}(\tau)}{S_{ij}(\tau)}, \tag{18}$$

where $f_{ij}(\tau)$ is the probability density function of $\mathcal{T}_{ij}$ and $S_{ij}(\tau)$ is the survival function which represents the probability that a susceptible individual j has not had an infectious contact with infected case i at time τ. From eq (18) the pdf of $\mathcal{T}_{ij}$ can be written as

$$f_{ij}(\tau) = h_{ij}(\tau)^{\delta_{ij}} S_{ij}(\tau). \tag{19}$$

where δ_{ij} is an indicator variable, taking the value 1 if individual j is infected by individual i and 0 otherwise. Hence each observed time τ_{ij}, the likelihood is given as

$$L_{ij}(\tau_{ij}) = h_{ij}(\tau_{ij})S_{ij}(\tau_{ij}). \tag{20}$$

4.2 Likelihood for a single transmission tree

Let $\mathcal{W}_j = \{i : t_i + \varepsilon_i < t_j \text{ and } C_{ij} = 1\}$ be the *exposure set* for individual j, which represents the set of all infectious cases i that individual j was exposed to while susceptible. Let v_j denote the infector of individual j, then for the other $i \in \mathcal{W}_j$ we get

$$\prod_{i \in \mathcal{W}_j; i \neq v_j} S(\tau_{ij}) = \prod_{i \in \mathcal{W}_j; i \neq v_j} e^{-\int_0^{l_i} h_{ij}(u)Y_{ij}(u)du}. \tag{21}$$

and thus, the j-th individual contribution to the likelihood from eq(20) is

$$L_j = h_{v_j j}(\tau_{v_j j}) \prod_{i \in \mathcal{W}_j} e^{-\int_0^{l_i} h_{ij}(u)Y_{ij}(u)du}. \tag{22}$$

Let $\mathbf{v} = (v_1, ...v_n)$ represent a transmission tree with a directed edge from v_j to j for each infected j, then the likelihood for a given tree $\mathbf{v}$ is given by

$$L_{\mathrm{v}}(\boldsymbol{\beta}, \boldsymbol{\theta}) = \prod_{j=1}^{n} \left[h_{v_j j}(t_j - t_{v_j} - \varepsilon_{v_j}; \boldsymbol{\beta}, \boldsymbol{\theta})^{1_{v_j \notin \{0,\infty\}}} \prod_{i \in \mathcal{W}_j} e^{-\int_0^{l_i} h_{ij}(\tau; \boldsymbol{\beta}, \boldsymbol{\theta})Y_{ij}(\tau)d\tau} \right], \tag{23}$$

where $v_j = 0$ if subject j is an exogenous infection and $v_j = \infty$ if subject j is not infected. Kenah et al. (2016) use parametric models for the hazard of infectious contact from i to j at time τ.

$$h_{ij}(\tau, \boldsymbol{\beta}, \boldsymbol{\theta}) = \exp\left(\boldsymbol{\beta}^\top X_{ij}(\tau)\right) h(\tau, \boldsymbol{\theta}), \tag{24}$$

where $X_{ij}(\tau)$ is a vector of covariates that may affect infectiousness, which may include the location of individuals, the health records etc. The coefficient vector $\boldsymbol{\beta}$ captures the effects of the covariates on the hazard of transmission. $h(\tau, \boldsymbol{\theta})$ is a baseline hazard function which captures the evolution of infectiousness over time in infectious individuals and is assumed to have continuous second derivative with respect to $\boldsymbol{\theta}$ to allow for maximum likelihood estimation.

4.3 Likelihood for multiple transmission trees

Let $\mathcal{V}$ be the set of all possible transmission trees $\mathbf{v}$ consistent with the observed data. When the actual transmission tree is unknown, then the likelihood is the sum over all possible transmission trees: $L(\boldsymbol{\beta}, \boldsymbol{\theta}) = \sum_{\mathrm{v} \in \mathcal{V}} L_{\mathrm{v}}(\boldsymbol{\beta}, \boldsymbol{\theta})$. A possible transmission tree is generated by choosing a particular v_j from the set $\mathcal{V}_j$ for a each sampled case j, where $\mathcal{V}_j$ denotes the set of possible infectors of case j. The likelihood can therefore be obtained by finding the product of the sum of the $\mathcal{V}_j$'s for each infected case j. This gives

$$L(\boldsymbol{\beta}, \boldsymbol{\theta}) = \prod_{j=1}^{n} \left[\sum_{\mathrm{i} \in \mathcal{V}_j} \left(h_{ij}(t_j - t_i - \varepsilon_i; \boldsymbol{\beta}, \boldsymbol{\theta})^{1_{v_j \notin \{0,\infty\}}} \prod_{i \in \mathcal{W}_j} e^{-\int_0^{l_i} h_{ij}(\tau; \boldsymbol{\beta}, \boldsymbol{\theta}) Y_{ij}(\tau) d\tau} \right) \right]. \tag{25}$$

Now, let Φ denote the phylogenetic tree obtained from the genetic sequences, $\mathbf{v}$ be a transmission tree, E denote the epidemiological data and genetic sequences, and the function $\mathrm{P}(\cdot)$ denote probability densities. The likelihood for $\boldsymbol{\beta}, \boldsymbol{\theta}$ if Φ and $\mathrm{v}'s$ are known is

$$L(\Phi, \mathrm{v}, E; \boldsymbol{\beta}, \boldsymbol{\theta}, \boldsymbol{\ell}, \boldsymbol{\kappa}) = \sum_{\mathrm{v} \in \mathcal{V}} P(\mathrm{v}, \Phi, E; \boldsymbol{\beta}, \boldsymbol{\theta}, \boldsymbol{\ell}, \boldsymbol{\kappa}) = \sum_{\mathrm{v} \in \mathcal{V}} P(\Phi | \mathrm{v}, E; \boldsymbol{\ell}, \boldsymbol{\kappa}) P(\mathrm{v}, E; \boldsymbol{\beta}, \boldsymbol{\theta}). \tag{26}$$

The component $P(\mathrm{v}, E; \boldsymbol{\beta}, \boldsymbol{\theta})$ is the likelihood in eq (23). Note that integrating eq (26) over the space of phylogenetic trees does not equal eq (25). The component $P(\Phi | \mathrm{v}, E; \boldsymbol{\ell}, \boldsymbol{\kappa})$ depends on the population dynamics (i.e. rate of growth on infections) and molecular evolution of the pathogen. The parameters of the models describing the population size, the growth rate, and the within-host evolution can be estimated using a Bayesian or frequentist approach as described in Section 3.

4.4 Accounting for covariates in Epidemiological models

A key component in inferring transmission trees for an outbreak is modelling the epidemiological process. The process describes how infections occur using information on the infection times, the sampling times, the number of contacts made, and the demographic data of sampled cases. Didelot et al. (2017) represent the probability of the full epidemiological process of a finished outbreak as the product of the generation time (time from a primary infection to any secondary infection) density, the sampling time (time from infection to sampling of a primary case) and the density of the number of secondary infections for each case in the outbreak. They model the generation time and sampling time using a Gamma distribution and

model the number of secondary infections for each case using a negative binomial distribution. The model proposed by Didelot et al. (2017) is very specific and does not easily allow the incorporation of other covariates such as age, location, number of person in household etc related to the risk of infection. Morelli et al. (2012) considered a similar approach to that of Didelot et al. (2017) where the probability of the full epidemiological process is represented a product of the generation time, sampling times and the risk of infection based on the distance between the location of an infection pair (infector and infectee). They describe the risk of infection between an infectious individual and susceptible individual as decreasing exponentially with the distance between the pair. They use an exponential transmission kernel as the density for the distance between an infection pair.

These approaches require some expertise in the field of epidemiology when modelling events that are believed to have an effect on the risk of infection. In addition, the proposed model may become complex when one decides to include more covariates to describe the epidemiological process. The survival framework presented by Kenah et al. (2016) naturally allows for the incorporation of covariates related to the risk of infection. The regression component of the model allows for a number of covariates which could result in more accurate estimates of the risk of infection. As stated earlier, we seek to incorporate spatial data in modelling transmissions and using the survival framework would serve as a good approach to incorporate spatial data and other factors such as age which could be important in assessing risk of infections in an outbreak.

5 Expectation-Maximization Algorithm for Parameter Estimation

5.1 Introduction

Even though Kenah et al. (2016) account for the uncertainty of not knowing the true transmission tree by listing all possible transmission trees, they use a single phylogenetic tree for the estimation of model parameters as shown in eq (26). The use of a single phylogenetic tree when finding the consistent transmission trees for the estimation of hazard of transmission parameters fails to account for the uncertainty associated with the choice of the phylogenetic tree. In their discussion, Kenah et al. (2016) note that their approach does not account for the phylogenetic uncertainty in the estimation of the hazard model parameters. They propose using Bayesian methods to reconcile the reconstruction of phylogenetic trees with possible transmission trees.

With the actual structure of the phylogeny not known, a selected phylogenetic tree may not be a true representation of the genetic evolution of the outbreak. Conclusions from approaches such as those proposed by Cottam et al. (2008), Didelot et al. (2014) and Kenah et al. (2016) may change noticeably if another reconstructed phylogenetic tree is used.

Also, a single phylogenetic tree may not necessarily agree with the observed epidemiological data and some assumptions may be needed in using the tree (Hall et al. 2015). Some assumptions may also be relaxed and parameter estimates may improve if all possible samples from the space of phylogenetic trees are used in the reconstruction of transmission tree and estimation of transmission parameters.

To properly treat the uncertainty associated with the unobserved transmission and phylogenetic trees, we propose using the expectation maximization (EM) algorithm in obtaining the maximum likelihood estimates (MLE) of parameters. For the likelihood in eq (26), we assume that the complete data set consists of the epidemiological data, the genetic data, the phylogenetic trees and the transmission trees. In our EM algorithm setup, the unobserved data are the phylogenetic trees and the transmission trees while the observed data are the epidemiological and genetic data. The wish is to find the MLE of the unknown parameters of the hazard model i.e.

$$\underset{\boldsymbol{\beta},\boldsymbol{\theta}}{\operatorname{argmax}} \sum_{\Phi,\mathrm{v}} L(\Phi, \mathrm{v}, E; \boldsymbol{\beta}, \boldsymbol{\theta}, \boldsymbol{\ell}, \boldsymbol{\kappa}),$$

where the likelihood is as described in eq (26). The EM algorithm allows us to better assess the uncertainty of parameter estimates.

5.2 Expectation Maximization (EM) Algorithm

In this section, we present the EM algorithm following Neal and Hinton (1998). They present an incremental algorithm where they show that M-step maximizes a negative free function with respect to the model parameters and the E-steps maximizes this function with respect to the distribution of the unobserved variables. The incremental variant of EM algorithm is shown empirically to give a faster convergence in a mixture estimation problem. Our motivation for presenting the EM algorithm in this way is to allow us to propose other distributions for the E-step of the EM algorithm if the distribution $P(\Phi|\mathrm{v}, E; \boldsymbol{\ell}, \boldsymbol{\kappa})$ is too hard to simulate from. In the sections below, we represent the log of the marginal distribution of the observed data as a sum of a lower bound of the log likelihood and the Kullback-Liebler divergence and

show how we can use the incremental algorithm for the maximization of the log likelihood function.

5.2.1 The Expectation Maximization (EM) Algorithm

Given observed variables $\mathbf{x}$, unobserved variables $\mathbf{z}$, and model parameters θ, the goal is to maximize the likelihood function

$$\mathcal{L}(\theta) = \log\, p(\mathbf{x};\theta) = log \int p(\mathbf{x},\mathbf{z};\theta)d\mathbf{z}, \tag{27}$$

over θ, where $p(\mathbf{x};\theta)$ is the marginal distribution of the observed variable $\mathbf{x}$ and $p(\mathbf{x},\mathbf{z};\theta)$ is the joint distribution of the observed and unobserved variables. The marginal distribution of the observed variable is written as

$$\begin{aligned} p(\mathbf{x};\theta) &= \int p(\mathbf{x},\mathbf{z};\theta)d\mathbf{z} \\ &= \int p(\mathbf{x}|\mathbf{z};\theta)q(\mathbf{z};\theta)d\mathbf{z} \\ &= \int h(\mathbf{z}|\mathbf{x};\theta)p(\mathbf{x};\theta)d\mathbf{z} \end{aligned}$$

The EM algorithm can be helpful when $p(\mathbf{x},\mathbf{z};\theta)$ can be written explicitly and $h(\mathbf{z}|\mathbf{x};\theta)$ or at least the expected value w.r.t. $\mathbf{z}|\mathbf{x}$ can be computed.

For any distribution $q(\mathbf{z})$, we can write the log likelihood of the marginal distribution of the observed variable as

$$\begin{aligned} \log\, p(\mathbf{x};\theta) &= \int [\log\, p(\mathbf{x};\theta)]q(\mathbf{z})d\mathbf{z} \quad \text{for any} \quad \int q(\mathbf{z})d\mathbf{z} = 1 \\ &= \int \log\left[\frac{p(\mathbf{x},\mathbf{z};\theta)}{p(\mathbf{z}|\mathbf{x};\theta)}\right]q(\mathbf{z})d\mathbf{z} \\ &= \int \log\, p(\mathbf{x},\mathbf{z};\theta)q(\mathbf{z})d\mathbf{z} - \int \log\, p(\mathbf{z}|\mathbf{x};\theta)q(\mathbf{z})d\mathbf{z} \\ &= \int \log\left(\frac{p(\mathbf{x},\mathbf{z};\theta)}{q(\mathbf{z})}\right)q(\mathbf{z})d\mathbf{z} - \int \log\left(\frac{p(\mathbf{z}|\mathbf{x};\theta)}{q(\mathbf{z})}\right)q(\mathbf{z})d\mathbf{z} \\ &= F(q;\theta) + KL(q|p(\mathbf{z}|\mathbf{x};\theta)) \end{aligned} \tag{28}$$

where $F(q;\theta)$ is a lower bound of the log likelihood and $KL(q|p(\mathbf{z}|\mathbf{x};\theta))$ is the Kullback-Liebler divergence with the following properties (proof shown below):

1. $KL(q|p(\mathbf{z}|\mathbf{x};\theta)) \geq 0$ & $KL(p(\mathbf{z}|\mathbf{x};\theta)|p(\mathbf{z}|\mathbf{x};\theta)) = 0$

   Hence, $p(\mathbf{z}|\mathbf{x};\theta)$ minimizes $KL(q|p(\mathbf{z}|\mathbf{x};\theta))$ w.r.t q.

2. $\log\, p(\mathbf{x};\theta) \geq F(q;\theta)\ \forall q\ \forall\theta$.

1. *Proof.* $KL(q|p(\mathbf{z}|\mathbf{x};\theta)) \geq 0$

   $KL(q|p(\mathbf{z}|\mathbf{x};\theta)) = \int -\log\,\left(\frac{p(\mathbf{z}|\mathbf{x};\theta)}{q(\mathbf{z})}\right)q(\mathbf{z})d\mathbf{z}$

   Jensen's inequality: If $\psi : \mathbb{R} \to \mathbb{R}$ is convex $\implies E_X(\psi(X)) \geq \psi(E_X(X))$

   $\psi(t) = -\log(t)$ is convex and $KL(q|p(\mathbf{z}|\mathbf{x};\theta)) = E_q[-\log\left(\frac{p(\mathbf{z}|\mathbf{x};\theta)}{q(\mathbf{z})}\right)]$

   We know that $KL(q|p(\mathbf{z}|\mathbf{x};\theta)) = E_q[-\log\,\left(\frac{p(\mathbf{z}|\mathbf{x};\theta)}{q(\mathbf{z})}\right)] \geq -\log[E_q(\frac{p}{q})] = -\log(1) = 0$

   Hence, $KL(q|p(\mathbf{z}|\mathbf{x};\theta)) \geq 0$.

   Now to show that $KL(p(\mathbf{z}|\mathbf{x};\theta)|p(\mathbf{z}|\mathbf{x};\theta)) = 0$, note that

$$KL(p(\mathbf{z}|\mathbf{x};\theta)|p(\mathbf{z}|\mathbf{x};\theta)) = \int -\log\,\left(\frac{p(\mathbf{z}|\mathbf{x};\theta)}{p(\mathbf{z}|\mathbf{x};\theta)}\right)p(\mathbf{z}|\mathbf{x};\theta)\,d\mathbf{z} \;=\; -\log(1) = 0\,.$$

   □

2. *Proof.* $\log\, p(\mathbf{x};\theta) \geq F(q;\theta)$ for any q

   $\log\, p(\mathbf{x};\theta) = F(q;\theta) + KL(q|p(\mathbf{z}|\mathbf{x};\theta))$

   Hence, $\log\, p(x;\theta) \geq F(q;\theta)$ since $KL(q|p(\mathbf{z}|\mathbf{x};\theta) \geq 0$. □

5.2.2 Applying variant to Standard EM algorithm

Following Neal and Hinton (1998), the EM algorithm can be thought of as alternating between optimizing the function $F(q;\theta)$ with respect to the distribution q and the parameter θ. Let θ^k be the current value of the parameter we want to estimate and let $q_k(\mathbf{z}) = p(\mathbf{z}|\mathbf{x};\theta^k)$, then from (28)

$$\log\, p(\mathbf{x};\theta^k) = F(q;\theta^k) + KL(q|p(\mathbf{z}|\mathbf{x};\theta^k)),\ \text{for any density function } q.$$

Now, since

$$\log\, p(\mathbf{x};\theta^k) \geq F(q,\theta^k),\ \text{for any density function } q. \tag{29}$$

and

$$\log p(\mathbf{x};\theta^k) = F(q_k,\theta^k)\,, \tag{30}$$

because $KL(q_k|p(\mathbf{z}|\mathbf{x};\theta^k)) = KL(q_k||q_k) = 0$, then

$$p(\mathbf{z}|\mathbf{x};\theta^k) = q_k(\mathbf{z}) = \arg\max_q F(q;\theta^k)\,. \tag{31}$$

For the standard EM Algorithm, the E-step involves finding the distribution of the unobserved variables, given the known values of the observed variable and the current estimate of the parameters and the M-step maximizes this function over the parameters of interest. Each iteration of the EM algorithm increases the log-likelihood function until the algorithm converges at the maximum of the log-likelihood function. Here, we show that the EM algorithm can be thought of as maximizing $F(q;\theta)$ by alternating between q and θ.

Given θ^{k-1}, let $q_{k-1}(\mathbf{z}) = p(\mathbf{z}|\mathbf{x};\theta^{k-1})$ and

$$\theta^{(k)} = \arg\max_\theta F(q_{k-1};\theta)\,. \tag{32}$$

The E-step finds q_{k-1} that maximizes the function $F(q;\theta^{k-1})$ and the M-step finds θ^k that maximizes the function $F(q_{k-1};\theta^{k-1})$.

We see that

$$\log(p(\mathbf{x};\theta^{(k)})) \geq F(q_{k-1};\theta^{(k)}) \geq F(q_{k-1};\theta^{(k-1)}) = \log(p(\mathbf{x};\theta^{(k-1)}))$$

(the first inequality follows from eq (29), the second from eq (32), and the last from eq (31)). Thus, the log-likelihood does not decrease.

This iterative algorithm of updating q and θ increases the likelihood at each stage and provides the maximum likelihood estimates for parameters. This variant of the EM algorithm is equivalent to the standard EM algorithm (Neal and Hinton, 1998) and we show below that maximizing the function $F(q_{k-1};\theta)$ is the same as the maximizing the expected log-likelihood in the EM algorithm.

Recall that

$$F(q;\theta) = \int \log\left(\frac{p(\mathbf{x},\mathbf{z};\theta)}{q(\mathbf{z})}\right) q(\mathbf{z})d\mathbf{z}$$

and $q_{k-1} = p(\mathbf{z}|\mathbf{x};\theta^{(k-1)})$ hence

$$\begin{aligned} F(q_{k-1};\theta) &= \int \log\left(\frac{p(\mathbf{x},\mathbf{z};\theta)}{p(\mathbf{z}|\mathbf{x};\theta^{(k-1)})}\right) p(\mathbf{z}|\mathbf{x};\theta^{(k-1)})d\mathbf{z} \\ &= \int \log(p(\mathbf{x},\mathbf{z};\theta))p(\mathbf{z}|\mathbf{x};\theta^{(k-1)})d\mathbf{z} - \int \log\Big(p(\mathbf{z}|\mathbf{x};\theta^{(k-1)})\Big)p(\mathbf{z}|\mathbf{x};\theta^{(k-1)})d\mathbf{z} \end{aligned}$$

The second term $\int \log\big(p(\mathbf{z}|\mathbf{x};\theta^{(k-1)})\big)p(\mathbf{z}|\mathbf{x};\theta^{(k-1)})d\mathbf{z}$ does not depend on θ, thus

$$\begin{aligned} \theta^{(k)} &= \arg\max_{\theta} F(q_{k-1};\theta) \\ &= \arg\max_{\theta} \int \log(p(\mathbf{x},\mathbf{z};\theta))p(\mathbf{z}|\mathbf{x};\theta^{(k-1)})d\mathbf{z} \\ &= \arg\max_{\theta} E_{\mathbf{z}|\mathbf{x};\theta^{(k-1)}} \log(p(\mathbf{x},\mathbf{z};\theta)) \,, \end{aligned}$$

which corresponds to the M-step in the usual formulation of the EM-algorithm.

5.3 Monte Carlo Methods

To estimate the conditional distribution of the phylogenetic and transmission trees given the genetic data, epidemiological data and parameter values, we employ Monte Carlo methods. Monte Carlo methods are used when calculating a quantity from a distribution is intractable. For example, we may be interested in computing the expected value, the median or the mode from a desired distribution but the estimator may not have a closed-form which makes it difficult to compute. Monte Carlo methods can be used to obtain samples from the distribution and numerically obtain an estimate of the quantity of interest. In this proposal we are interested in approximating $E_{\mathbf{z}|\mathbf{x};\theta^{(k-1)}} \log(p(\mathbf{x},\mathbf{z};\theta))$.

5.3.1 Importance Sampling

Importance Sampling (IS) approximates the expectation of a function with respect to a desired distribution using samples that have been generated from a different distribution. The appeal for Importance sampling lies in a simple probability result. Let $f(x)$ be a probability density of a random variable X and suppose

we wish to compute an expectation $E_f[h(X)]$, with

$$E_f[h(X)] = \int h(x)f(x)dx.$$

For any probability density $q(x)$ that satisfies $q(x) > 0$ whenever $h(x)f(x) \neq 0$, one has

$$E_f[h(X)] = E_q[w(X)f(X)]$$

where $w(x) = h(x)/q(x)$ and now $E_q[\cdot]$ denotes the expectation with respect to the distribution q. The method of IS involves independent draws $x^{(1)}, ..., x^{(m)}$ from the distribution $q(x)$ and approximating $E_f[h(X)]$ by

$$E_f[h(X)] \approx \frac{1}{m}\sum_{j=1}^{m} w(x^{(j)})f(x^{(j)}).$$

It follows from the strong law of large numbers that $\frac{1}{m}\sum_{j=1}^{m} w(x^{(j)})f(x^{(j)}) \to E_f[h(X)]$ as $n \to \infty$ almost surely. The approximation accuracy offered by IS depends critically on the choice of the trial density $q(x)$. For IS to be accurate with a limited number of draws the variance must be small, which requires $q(x)$ to be approximately proportional to $f(x)$ for most x.

Important sampling is appealing in our setting given the fact our current setting takes new samples from the MCMC for the phylogenetic tree and transmission tree for each iteration of the EM algorithm. This approach can be computationally expensive. Using IS, we can recycle the chains from the previous iteration since each iteration increases the likelihood and the sample from the previous iteration is close enough to the target for the current iteration. We consider this approach in our further works.

5.3.2 Markov Chain Monte Carlo methods

Markov Chain Monte Carlo (MCMC) methods are used to simulate independent and identically distributed samples from a target distribution. The MCMC samples from a Markov chain with invariant distribution for the target. It is often easier to sample from such a chain than to obtain samples from the target distribution. The procedure combines two properties; Monte-Carlo which involves estimating the properties of a target distribution using random samples from the distribution and Markov chain property where samples are drawn sequentially, with the distribution of the sampled draws depending

on the last value drawn. Each random sample is used as a stepping stone to generate the next random sample. These simulation methods produce a sequence X_1, ... , X_N which is a Markov chain. i.e., such that the distribution of X_i given the past $\{X_{i-1}, ..., X_1\}$ only depends on X_{i-1}.

Commonly used MCMC methods include the Metropolis Hastings algorithm and the Gibbs Sampler. The Metropolis Hasting process requires a conditional proposal distribution, with density $q(y|x)$ in the generation of samples. For a target density h, the process is described as follows:

Given the current state $x^{(t)}$

1. Generate $Y_t \sim q(y|x^{(t)})$
2. Take

$$X^{(t+1)} = \begin{cases} Y_t & \text{with probability } \rho(x^{(t)}, Y_t), \\ x^{(t)} & \text{with probability } 1 - \rho(x^{(t)}, Y_t), \end{cases}$$

where

$$\rho(x, y) = \left\{ \frac{f(y)q(x|y)}{f(x)q(y|x)}, 1 \right\}.$$

The Metropolis-Hastings algorithm can be implemented in practice when $q(\cdot|x)$ is easy to simulate from and is either explicitly available (up to a multiplicative constant *independent* of x) or *symmetric*; that is, such that $q(x|y) = q(y|x)$.

Gibbs sampler is a special case of the Metropolis Hastings algorithm which is used to simulate from complex joint distributions when we have access to the full conditionals. Simulation is done by reducing multi-dimensional problems to lower-dimensional problems. The entire vector is subdivided into smaller sub-vectors before randomly sampling from conditional distributions. Suppose the random variable $\mathbf{X} = (X_1, .., X_p)$, where the X_i's are either uni-or multidimensional and that we can simulate from the corresponding univariate conditional densities $f_1, .., f_p$, that is, we can simulate

$$X_i|x_1, x_2, ..., x_{i-1}, x_{i+1}, ..., x_p \sim f(x_i|x_1, x_2, ..., x_{i-1}, x_{i+1}, ..., x_p)$$

for $i = 1, 2, ..., p$. The associated Gibbs sampling algorithm is given by the following transition from $X^{(t)}$ to $X^{(t+1)}$:

Given $\mathbf{x}^{(t)} = (x_1^{(t)}, ..., x_p^{(t)})$ generate

1. $X_1^{(t+1)} \sim f_1(x_1|x_2^{(t)}, ..., x_p^{(t)})$
2. $X_2^{(t+1)} \sim f_2(x_2|x_1^{(t+1)}, x_3^{(t)}, ..., x_p^{(t)})$

   $\vdots$

p. $X_p^{(t+1)} \sim f_p(x_p|x_1^{(t+1)}, ..., x_{p-1}^{(t+1)})$

The densities $f_1, ..., f_p$ are called full conditionals, and a particular feature of the Gibbs sampler is that these are the only densities used for simulation. Thus, even in a high-dimensional problem, all of the simulations maybe univariate, which is usually an advantage (Robert and Casella, 1999).

5.4 Monte Carlo - EM

The Expectation Maximization algorithm is used to obtain the maximum likelihood estimates for models that yield analytically formidable likelihood equations (Levine and Casella, 2001). The procedure involves two steps where we first obtain the expectation of the log-likelihood (E-step) and maximize this expectation over the parameters of interest (M-step). Some methods have been developed to help in the estimation of parameters in the case where the E-step in the EM algorithm is not closed formed or analytically intractable. The Monte Carlo-EM algorithm which was introduced by Wei and Tanner (1990) is one of such methods. The procedure makes use of Monte Carlo simulations in approximating the expectation of the log-likelihood.

The basic idea behind the EM algorithm is to augment the observed data x by the latent quantity z. We assume that, given both x and z, computation and maximization of the expectation of the complete-data log-likelihood $\log f(x, z; \theta)$ is straight forward (Wei and Tanner, 1990). The EM algorithm iterates between the calculation of the expected complete-data likelihood

$$Q(\theta; \theta^{k-1}) = \int_z log f(x, z|\theta) g(z|x, \theta^{k-1}) dz \tag{33}$$

and the maximization of the $Q(\theta; \theta^{k-1})$ over θ to obtain the update θ^k. In the situation were the E-step is

intractable, the Monte Carlo method can be used to approximate the integration in eq (33). The approach is described below:

Given the current approximation to the maximizer θ^{k-1},

1. generate a sample $z^1, ..., z^m$ from the distribution $g(z|y, \theta^{k-1})$, this expectation may be estimated by the Monte Carlo sum

$$Q_m(\theta; \theta^{k-1}) = \frac{1}{m} \sum_{j=1}^{m} \log f(x, z^j|\theta) \tag{34}$$

   where the subscript m denotes the dependence of this estimator on the MC sample size.

2. Maximize the right side of eq (34).

Samples for the approximation of the E-step can be obtained using Monte Carlo methods or Markov Chain Monte Carlo (MCMC) techniques such as the Gibbs Sampler and Metropolis-Hastings algorithm. The chains from these methods could also be reused via importance sampling to obtain our samples for each iteration. We consider these techniques in the estimation of hazard parameters as shown in section 5.6.

5.5 Variational approximation EM Algorithm

Another approximation that is used in the case where the integration over the latent variable is computationally intractable is the Variational approximation EM algorithm. This variant of the EM algorithm may be useful for our approach, especially in the situation where E-step of the EM algorithm is not trivial and we will need to approximate the distribution. From the review of the EM algorithm in section 5.2, the variational EM seeks to find $q(\mathbf{z})$ which maximizes $F(q; \theta)$ when probability measure q is restricted over a certain class. We wish to find

$$\tilde{q}_{k-1} = \arg\max_{\tilde{q} \in \tilde{Q}} F(q; \theta^{k-1}).$$

for some set of distributions $\tilde{Q}$. A common set of distributions on which we can restrict the search is

$$\tilde{Q} = \left\{ \tilde{q}(\mathbf{z}) = \prod_j \tilde{q}_j(z_j) \right\}. \tag{35}$$

In other words, the missing variables are independent of each other (Tzikas et al., 2008). From the setup of the EM algorithm in the previous sections, we see that

$$\begin{aligned} \log p(x;\theta) &= F(q;\theta) + KL(q|p(\boldsymbol{z}|\boldsymbol{x};\theta)) \ \forall q \ \in \mathcal{M}_1 \\ &= F(\tilde{q};\theta) + KL(\tilde{q}|p(\boldsymbol{z}|\boldsymbol{x};\theta)) \ \forall \tilde{q} \ \in \tilde{Q} \end{aligned}$$

From eq(35) we have that

$$\begin{aligned} F(\tilde{q};\theta) &= \int \log \left(\frac{p(x,z;\theta)}{\tilde{q}(z)} \right) \tilde{q}(z) dz \\ &= \int \log \left(p(x,z;\theta) \right) \prod_{j=1}^{d} \tilde{q}_j(z_j) dz_j - \int \left(\sum_{j=1}^{d} \log \tilde{q}_j(z_j) \right) \prod_{j=1}^{d} \tilde{q}_j(z_j) dz_j \\ &= \int \log \left(p_j(x,z_j;\theta) \right) \tilde{q}(z_j) dz_j - \sum_{l=1}^{d} \int \log \tilde{q}_l(z_l) \tilde{q}_l(z_l) dz_l \\ &= \int \log \left(\frac{p_j(x,z_j;\theta)}{\tilde{q}_j(z_j)} \right) \tilde{q}_j(z_j) dz_j - \sum_{l \neq j}^{d} \int \log \tilde{q}_l(z_l) \tilde{q}_l(z_l) dz_l \end{aligned}$$

where $\log p_j(x,z_j;\theta) = \int \log p(x,z;\theta) \prod_{l \neq j}^{d} \tilde{q}_l(z_l) dz_l$. This implies that

$$F(\tilde{q};\theta) = -KL(\tilde{q}_j|p_j) - \sum_{l \neq j}^{d} \int \log \tilde{q}_l(z_l) \tilde{q}_l(z_l) dz_l \quad \text{for any } 1 \leq j \leq d$$

Hence the optimal $\tilde{q}_j's$ satisfy

$$\tilde{q}_j(z_j) = p_j(\mathbf{x}, z_j;\theta) \text{ for each } 1 \leq j \leq d. \tag{36}$$

It is key to note that these conditions are sufficient, but do not provide a closed for expression for the optimal $\tilde{q}$, since the right hand side of eq (36) depends on all the other components $\tilde{q}_\ell, \ \ell \neq j$. This to find a set of simultaneous solutions to the d equations in eq (36), one may need to iterate.

5.6 Accounting for uncertainty in phylogenetic trees for a frequentist framework

Our interest is in estimating the survival model parameters from eq (26). We propose a Monte Carlo-EM algorithm approach where samples for the phylogenetic tree and transmission tree are obtained using the model proposed by Klinkenberg et al. (2017).

5.6.1 Monte Carlo-EM algorithm for Transmission model

We begin by defining the notations for our model. Let:

- E: the epidemiological data,
- I: the set of infectious times,
- G: genetic data from pathogens,
- S: the sampling times of the genetic data,
- Φ: the unobserved phylogenetic tree, and
- v: the unobserved transmission tree.

The full data here is denoted by (Φ, v, S, E, I, G) and the observed data is denoted by (S,E,I,G). The vector of parameters $\mathbf{\Theta}$ comprises of the hazard function parameter (λ), the hazard regression parameter (β) and the evolution/molecular clock parameters (ℓ, κ). The likelihood for the observed data is given as

$$L(S, E, I, G; \mathbf{\Theta}) = \sum_{(\Phi, v)} L(\Phi, v, S, E, I, G; \mathbf{\Theta}). \tag{37}$$

We use the EM algorithm to maximize eq (37) with respect to $\mathbf{\Theta}$. The E-step in the algorithm computes the expectation of the phylogenetic tree and transmission tree given the observed data and parameters.

$$\begin{aligned} H_{\mathbf{\Theta}^k}(\mathbf{\Theta}) &= E_{\Phi, v|SEIG; \mathbf{\Theta}^k}\left[log\ L(\Phi, v, S, E, I, G; \mathbf{\Theta})\right] \\ &= \int \log\ L(\Phi, vS, E, I, G; \mathbf{\Theta}) dP(\Phi, v|S, E, I, G; \mathbf{\Theta}^k). \end{aligned} \tag{38}$$

The conditional distribution of the unobserved phylogenetic and transmission trees given the observed data (S, E, I, G) and current parameters is denoted by $P(\Phi, v|S, E, I, G, \mathbf{\Theta})$. Since $P(\Phi, v|S, E, I, G, \mathbf{\Theta})$

is generally analytically intractable, we approximate $H_{\mathbf{\Theta}^k}(\mathbf{\Theta})$ using Monte Carlo.

$$H_{\mathbf{\Theta}^k}(\mathbf{\Theta}) \approx \frac{1}{M}\sum_{j=1}^{M} log\ L(\Phi^j, v^j, S, E, I, G; \mathbf{\Theta}), \tag{39}$$

where (Φ^j, v^j) are samples of phylogenetic and transmission trees obtained from $P(\Phi, v|S, E, I, G; \mathbf{\Theta}^k)$ using Metropolis Hastings.

To sample from $P(\Phi, v|S, E, I, G; \mathbf{\Theta}^k)$, we follow the innovation proposal of Klinkenberg et al. (2017). We have

$$\begin{aligned} P(\Phi, v|S, E, I, G; \mathbf{\Theta}) &\propto P(G|\Phi, S, E, I, v; \mathbf{\Theta})P(S, E, I, \Phi, v; \mathbf{\Theta}) \\ &\propto P(G|\Phi; \mathbf{\Theta})P(\Phi|S, E, I, v; \mathbf{\Theta})P(S, E, I, v; \mathbf{\Theta}) \\ &\propto P(G|\Phi; \mathbf{\Theta})P(\Phi|S, I, v; \mathbf{\Theta})P(S, E, I, v; \mathbf{\Theta}) \end{aligned} \tag{40}$$

$P(G|\Phi, S, E, I, v; \mathbf{\Theta})$ is the mutation model which reduces to $P(G|\Phi; \mathbf{\Theta})$ since this is the model for the genetic data and this does not make use of the epidemiological data, infectious times, sampling times and transmission trees. $P(\Phi|S, E, I, v; \mathbf{\Theta})$ is the coalescent model which reduces to $P(\Phi|S, I, v; \mathbf{\Theta})$ since this does not depend on the epidemiological data.

5.6.2 The models and likelihood

We describe the likelihood for each component of the model used in generating the samples. The first component is the mutation model where a single fixed mutation rate μ is used for all sites, with mutation resulting any of the four nucleotides with equal probability (Jukes-Cantor). For the given phylogenetic tree Φ, the likelihood is given as:

$$P(G|\Phi, \mu) = \prod_{loci} \sum_{\{A,C,G,T\}^{3e-1}} \prod_{x} \left(\frac{1}{4} - \frac{1}{4}exp(-\mu(t_x - t_{v_x}))\right)^{I_{ty}(1-N)} \cdot \left(\frac{1}{4} + \frac{3}{4}exp(-\mu(t_x - t_{v_x}))\right)^{(1-I_{rty})(1-N)} \tag{41}$$

This likelihood is calculated using Felsenstein's pruning algorithm (Felsenstein, 1981).

The second component is the coalescent model which accounts for coalescent process within the host. The model assumes that each host i has its own mini phylogenetic tree Φ_i, where the tips represent the transmission and sampling events, and the root represents the time of infection for host i. The likelihood

for the coalescent model is therefore the product of each host's likelihoods:

$$P(\Phi|S,I,v,r) = \prod_i P(\Phi_i|S_i,I,v,r)$$

where r is the parameter describing the within-host dynamics. The likelihood for each host's mini-tree is dependent on all infection times and infectors, and can be written as

$$P(\Phi_i|S_i,I,v,r) = \exp\left(-\int_0^\infty \binom{L_i(\tau)}{2}\frac{1}{w(\tau,r)}d\tau\right)\prod_{x|x\in\Phi_i\cap n<x<2n}\frac{1}{w(\tau_x,r)}, \tag{42}$$

where $w(\tau,r) = r\tau$ denotes a linearly increasing within-host pathogen population size at (forward) time τ since infection of the host. When we go backwards in time, coalescence between any pair of lineages within a host takes place at rate $1/w(\tau,r)$. $L_i(\tau)$ denotes the number of lineages in host i at time τ since infection:

$$L_i(\tau) = 1 + \sum_{x|x\in\Phi_i\cap n<x<2n} u(\tau-\tau_x) - \sum_{x|x\in\Phi_i\cap x\geq 2n} u(\tau-\tau_x) - u(\tau-\tau_i),$$

where $u(\tau) = 0$ if $\tau < 0$, and $u(\tau) = 1$ if $\tau \geq 0$.

The third component $P(S,E,I,v;\lambda,\beta)$ is the likelihood for the transmission tree, sampling times, infection times and epidemiological data. We employ the likelihood used by Kenah et al.(2016) which is described in eq (23).

5.6.3 Algorithm

1. Initialize $\mathbf{\Theta}^k$.

2. Generate $(\Phi^1,v^1),...(\Phi^M,v^M) \sim P(\Phi,v|S,E,I,G;\mathbf{\Theta}^k)$ via an MCMC algorithm.

3. E-step: Estimate $H_{\mathbf{\Theta}^k}(\mathbf{\Theta}) = E_{\Phi,v|SEIG;\mathbf{\Theta}^k}\left[log\ L(\Phi,v,S,E,I,G;\mathbf{\Theta})\right]$ by

$$\tilde{H}_{\mathbf{\Theta}^k}(\mathbf{\Theta}) = \frac{1}{M}\sum_{j=1}^{M} log\ L(\Phi^j,v^j,S,E,I,G;\mathbf{\Theta}^k).$$

4. M-step: Maximize $\tilde{H}_{\boldsymbol{\Theta}^k}(\boldsymbol{\Theta})$ to obtain $\boldsymbol{\Theta}^{k+1}$.

$$\boldsymbol{\Theta}^{k+1} = \underset{\boldsymbol{\Theta}}{\operatorname{argmax}} \quad H_{\boldsymbol{\Theta}^k}(\boldsymbol{\Theta}).$$

5. Set $\boldsymbol{\Theta}^k = \boldsymbol{\Theta}^{k+1}$.

6. Repeat Step 2 though 5 until convergence.

6 Implementation

Our proposed method combines processes implemented by Kenah et al. (2016) where the epidemiological data is used to list all possible transmissions trees based on the infectious and recovery times of cases in the data, and by Klinkenberg et al. (2017) where the genetic data for these sampled cases is then used to generate phylogenetic trees according to the coalescent model for each sampled transmission tree. In the generation of new phylogenetic trees, we implement proposal steps by Klinkenberg et al. (2017) where mini-phylogenetic trees for hosts are updated when changes occur in transmission tree. The proposal is done by comparing the infectors of hosts (sampled cases) in a new transmission tree to the initialized transmission tree and updating the phylogenetic tree based on cases where the infector of the host changes.

We initialize the MCMC chain with initial values for the mutation rate (μ), the slope parameter of the coalescent model (r), the hazard function parameters (β, λ), a randomly chosen transmission tree and the generated phylogenetic tree.

Each MCMC iteration cycle to generate data for the EM algorithm starts with a selection of a transmission tree and generation of the corresponding phylogenetic tree. These are then updated by proposing a new transmission tree from the set of all possible transmission trees and then obtaining its corresponding phylogenetic tree ($v^{'}, \Phi^{'}$). The proposal is accepted with Metropolis-Hastings probability α,

$$\alpha = min\left(1, \frac{P(G|\Phi^{'};\boldsymbol{\Theta})P(\Phi^{'}|S,I,v^{'};\boldsymbol{\Theta})P(S,E,I,v^{'};\boldsymbol{\Theta})}{P(G|\Phi;\boldsymbol{\Theta})P(\Phi|S,I,v;\boldsymbol{\Theta})P(S,E,I,v;\boldsymbol{\Theta})}\right) \tag{43}$$

Numerical methods are used to obtain parameter estimates via R. The initial draft of our procedure is implemented as a package in R (PhyTranSurv), which can be downloaded from GitHub (https://github.com/msalibian/PhyTranSurv).

7 Application

We apply the Monte Carlo-EM algorithm to estimate the parameters of the hazard function using genetic and epidemiological data on farms in the "Darlington cluster" of the 2001 foot and mouth disease in UK based (Morelli et al. 2012). The epidemiological data on 12 farms is shown Table 1 below. We have the farm names, location coordinates, sampling times (Tobs), time of culling (Tend) and the number of days since infectiousness (Di). They assume a latency period of 2-16 days.

Premise	Latitude	Longitude	Tobs	Tend	Di	Tinf
K	54.62088	-1.79246	41	44	1	40
L	54.60621	-1.75728	50	50	2	48
O	54.56211	-1.82062	54	55	2	52
F	54.58591	-1.70998	63	64	4	59
E	54.60676	-1.65931	63	64	7	56
C	54.62392	-1.61046	60	60	3	57
P	54.52640	-1.67180	70	71	4	66
M	54.58693	-1.77893	87	91	5	82
G	54.57489	-1.70432	84	84	4	80
D	54.59905	-1.78757	81	81	1	80
I	54.56362	-1.68972	98	98	3	95
J	54.57230	-1.64390	104	105	2	102

Table 1: Table with epidemiological information for the 12 farms

7.1 Hazard and Survival Function

To compute the likelihood in eq (31), we first obtain the probability $P(S, E, I, v; \mathbf{\Theta})$ which is given in eq (23). Kenah et al. (2016) propose a log-logistic contact interval distribution for the hazard function as defined below:

$$h(\tau, \lambda, \gamma) = \frac{\gamma \lambda^{\gamma} \tau^{\gamma-1}}{1 + (\lambda\tau)^{\gamma}} \tag{44}$$

where λ is the rate parameter and γ is the shape parameter. The survival function is given as $S(\tau, \lambda, \gamma) = (1 + (\lambda\tau)^{\gamma})^{-1}$. Without a phylogenetic tree, the likelihood contribution for each infected farm j, except

the index farm, overall possible infection times is given as:

$$P(S, E, I, v; \boldsymbol{\lambda}, \boldsymbol{\gamma}) = \sum_{t=t_j^*-16}^{t_j^*-2} \left[\sum_{i \in \mathcal{V}_j(t)} h(t - t_i - \varepsilon_i, \lambda, \gamma) \prod_{i \in \mathcal{W}_j(t)} S(min(l_i, t - t_i - \varepsilon_i), \lambda, \gamma) \right], \tag{45}$$

We combine eq (41), eq (42), and eq (45) to obtain the proposal distribution of the phylogenetic and transmission trees given the observed data in eq (40). We sum over all phylogenetic and transmission trees and maximize to obtain the estimates of the hazard transmission function.

7.2 Results

We present the results for the estimates considering various MCMC samples for different number of iterations of the EM algorithms. We initialize the EM algorithm with different values for the λ and γ and fix the values for the mutation rate ($\mu = 0.00005$) and the slope ($r = 0.5$), of the coalescent model.

	Point estimate		
Parameter	10000	25000	50000
λ	0.151	0.144	0.149
γ	3.065	2.828	2.846

Table 2: Estimates for varying MCMC samples within initial values of $\lambda = 0.30$ $\gamma = 1.2$ and 10 iterations (sample runs), 50 iterations for optimization.

	Point estimate		
Parameter	10000	25000	50000
λ	0.125	0.136	0.123
γ	1.989	2.177	1.921

Table 3: Estimates for varying MCMC samples within initial values of $\lambda = 0.30$ $\gamma = 1.2$ and 10 iterations (sample runs), 100 iterations for optimization

The estimates obtained by Kenah et al. (2016) before and after incorporating the phylogenetic tree were $\lambda = 0.132, \gamma = 2.486$ and $\lambda = 0.125, \gamma = 2.233$, respectively. The estimates we obtain are similar to the those of Kenah et al. (2016).

7.3 Simulation Study

We also implement our proposed approach using a simulation study. We simulated an outbreak of 10 cases. The sample size of 10 cases is to enable us enumerate all possible transmission trees that can be obtained from the epidemiological data.

The hazard of infectious contact from i to j at infectious age τ of i is given as

$$h_{ij}(\tau, \beta) = exp(\beta_{inf} X_i + \beta_{sus} X_j)\lambda_0.$$

where X_i and X_j are binary covariates that affect infectiousness and susceptibility, respectively and β_{inf} and β_{sus} are the hazard model parameters which are independently chosen from a uniform distribution on (-1,1). In all the simulations, the baseline hazard, λ_0 is set to one. The infectious periods are independent exponential random variables with mean one. After generating the epidemiological data, we simulate the genetic data using the infection and sampling times from the simulated outbreak in R. This is done using the proposed method in *Phybreak* (Klinkenberg et al. 2017).

Simulation of the outbreak and the epidemiological data are generated in python where as the generation of the genetic data and the estimation of model parameters is done in R. Results are presented below.

	Point estimate		
Parameter	10000	25000	50000
β_{inf}	-0.37	0.96	0.89
β_{sus}	-0.54	-0.18	-0.17
$ln\lambda$	-1.70	-0.82	-0.80

Table 4: Estimates for varying MCMC samples with true values of $\beta_{inf} = -1$, $\beta_{sus} = -1$, and $ln\lambda = -2.2$ and 5 EM iterations (sample runs), 100 iterations for optimization

	Point estimate		
Parameter	10000	25000	50000
β_{inf}	-0.37	0.91	0.91
β_{sus}	-0.54	-0.17	-0.17
$ln\lambda$	-1.70	-0.81	-0.81

Table 5: Estimates for varying MCMC samples with true values of $\beta_{inf} = -1$, $\beta_{sus} = -1$, and $ln\lambda = -2.2$ and 10 EM iterations (sample runs), 100 iterations for optimization

Estimates obtained from our simulation study are quite different from the true value and this could be due to the limitation of a small sample size. In our process, we list all possible transmission trees and perform the MCMC by randomly selecting transmission trees instead of rearranging connections in the tree. We seek to develop an approach that allows for a larger sample size in estimating the hazard model parameters.

Another reason for the large errors in the estimates is due to the small signal to noise ratio. A quick simulation study showed us that the parameter λ masks the effect of the β on the hazard function for the small sample size. Also, there are a number of parameters to account for in the proposed approach and having a small sample size can create a computational problem.

The simulation result, however, is proof of concept and our next steps include developing an approach where we update trees along side the phylogenetic trees in the MCMC for the EM algorithm.

8 Further work

The next steps of this approach is to develop a tractable theoretical framework or at least a non Monte-Carlo method for maximizing the expected value in (37). We seek to develop an approach that allows for sampling a large data set and makes it possible to list all possible trees associated with the data.

We also seek to incorporate the spatial data into the model to help improve the estimates of the parameters of the hazard function. The spatial component can be incorporated as a covariate in the cox proportional hazard in eq (24). We will look at different distance kernels and assess how the correlations between locations will affect parameter estimates.

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